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# Instance Recognition of Machining Features in B-rep CAD Models

*Three AI approaches: self-supervised point-cloud encoders, gradient-boosted  
face-pair classification, and a tool-using LLM agent*

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Final Year Internship Report

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# Abstract

A machined part is designed twice: once as a shape, in a computer-aided design (CAD) system, and once as a process, in a computer-aided manufacturing (CAM) system. One important step between the two, recognising which faces of the finished solid together form one machining feature, is still done by hand, by an experienced engineer, for every new part. This internship at Siemens AG addressed the *instance* half of that step on real industrial parts inside Siemens NX: given a boundary representation, decide which faces belong to the same feature. Two things make the problem hard. Labelled real parts are scarce, because companies do not publish them; and what counts as *one* feature is not settled, even among the people who label them.

Three approaches were built and measured, each a different bet on where the knowledge should come from. The first learns it from a million generated parts: a self-supervised point-cloud encoder, continually pretrained on a corpus whose labels were read off the construction history of the programs that made it, rather than annotated. The second engineers it from a few hundred real ones: a gradient-boosted classifier over face pairs, developed through a sequential set of experiments and carried to the deployment path of the product. The third trains nothing at all: a large language model (LLM) agent that reads the labelling conventions as text and explores the part inside Siemens NX with exact geometric queries, colouring the faces it decides belong together.

On generated parts, and when labels were scarce, pretraining was worth ten times as many labelled parts. On real parts the classifier rose from 51 to 76 % exact instance recognition, and a ledger of one-change experiments attributed the gain to label quality first, data volume second and engineered features a distant third. Starting from an unlabelled part, the agent reaches 77 % exact instance recognition on held-out parts of up to 80 faces, against 61 % for the trained classifier on 117 of the same parts and 74 % for the geometric rules it follows. All three ran into the same wall, and it is not the model: the consistency of what counts as one feature bounds what any of them can be measured to achieve. The report shows that the labelling convention is a design decision that can be tested on fixed folds like a model change, and argues that it is the lever the field under-uses.

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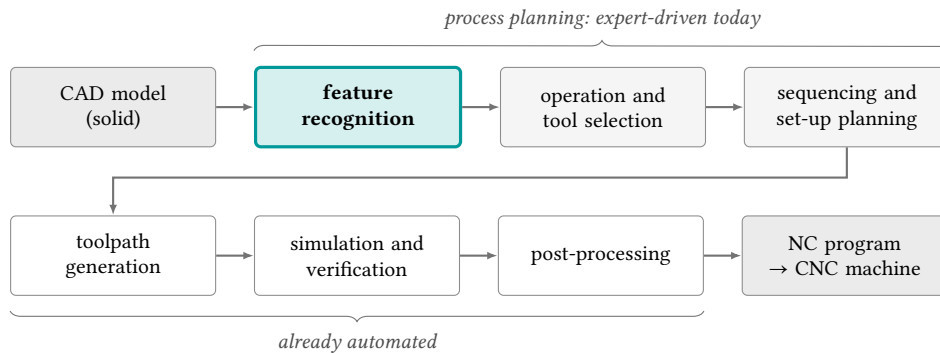
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# 1 Introduction

## 1.1 From CAD to CAM

A machined metal part is designed twice: once as a *shape*, in a computer-aided design (CAD) system such as Siemens NX, an exact geometric model of the finished component; and once as a *process*, in a computer-aided manufacturing (CAM) system, which decides how a machine tool will cut that shape out of a block of metal and emits the numerical control (NC) program that drives a computer numerical control (CNC) mill or lathe. Between the two sits *process planning* (Figure 1.1): which machining operations produce the part, which cutting tool each uses, in which order, and how the part is clamped for each set-up. Toolpath generation, simulation and post-processing are automated in every commercial CAM system; the decisions before them are made by an experienced engineer, part by part, and they take a long time.



**Figure 1.1:** The chain from a CAD solid to a machined part. The three white stages of the bottom row are already automated; machining-feature recognition is the first manual stage and gates the other two, which both consume feature *instances*.

The engineer’s first act is to *see* the solid in manufacturing terms: this cylindrical face and the conical face beneath it are one drilled and countersunk hole; those five faces are one closed pocket; that thin band is a chamfer on its rim. Commercial CAM systems have shipped rule-based recognisers for decades, and they work on the parts their rules anticipate. No complete and general rule set exists, and *interacting* features are precisely the cases hand-written rules cannot cover [1]: a pocket whose wall a later slot has cut away no longer looks like its textbook template. A learned recogniser is retrained from examples instead of re-programmed.

## 1.2 Machining-feature recognition

The stage this report attacks is *machining-feature recognition* (MFR): recovering, from finished geometry, the holes, pockets, slots, bosses, chamfers and fillets that map onto machining operations [1]. Industrial kernels store a part as a *boundary representation* (B-rep): trimmed parametric surface patches, the *faces*, joined by an explicit topology recording which faces share

**Table 1.1:** The three approaches of this report, in the order they were built.

| Approach                                   | Where the knowledge comes from                                            | Representation and scorer                                                                                                                                   |
|--------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 deep learning at scale                   | a million generated parts, labelled by their own construction history     | whole-part point cloud; Point Transformer V3 encoder pretrained on Z2C-Seg, face-affinity decoder                                                           |
| 2 classical machine learning on real parts | a few hundred hand-labelled real parts                                    | FeatureFox’s face-pair attributes and calibrated gradient boosting with a global threshold, extended to non-adjacent pairs and re-tuned on Siemens NX parts |
| 3 LLM agent inside Siemens NX              | the labelling conventions as text, and the part itself, queried on demand | no trained model: an agent with exact geometric tools labels the whole part, one instance at a time                                                         |

which edges. A feature is therefore a *set of faces*, and MFR splits into two sub-problems that fail differently: *semantic* labelling, which names the feature type of a face, and *instance* grouping, which decides which faces form one feature. The instance half is the operationally relevant one, because a CAM system selects a tool and an approach direction for a face *set*: a feature split in two is machined twice, and two features welded into one are machined wrongly or not at all. This report is about the instance half.

### 1.3 Objective and problem statement

The main objective of the internship, carried out at Siemens AG in the FT RPD AMC DMT-DE team (Section 3.1), was to recognise machining-feature instances on real Siemens NX parts: to decide which faces of a part belong to the same feature, so that Siemens NX CAM can treat them as one machinable entity. Four secondary outcomes were expected by its second half: a deployable instance-recognition model for real Siemens NX parts, extending the team’s FeatureFox recogniser; a measured answer to whether a large language model (LLM) agent inside Siemens NX, which trains no model and reads the labelling conventions and their measured rates, can recognise instances on those parts; Z2C-Seg (Zero-to-CAD segmentation), generated parts labelled by their construction history, and what pretraining on it buys; and labelled real-part corpora under a written convention. The two approaches on real parts share one metric, *exact instance recognition*: the share of ground-truth instances reproduced with exactly the same face set, averaged per part (Section 4.2); the approach on generated parts reports instance F1.

Two properties of the setting shaped the work more than any architectural choice. **The ground truth is itself uncertain**: what counts as *one* feature is not settled, and this turned out to be the largest difficulty of the internship. **The data is expensive and small**: the corpora are counted in hundreds of parts because each is hand-labelled inside Siemens NX, so label consistency matters more than label volume, and a handful of outliers can move the metric.

## **1.4 Three approaches**

The internship explored three approaches to the same problem, named in Table 1.1. Each is a different bet on where the knowledge that groups faces into features should come from: learned from a million generated parts, engineered from a few hundred real ones, or reasoned out by a model that can query the part and read the labelling conventions as text.

## **1.5 Report organisation**

Chapter 2 reviews the state of the art and closes on the gaps that motivate the three approaches; Chapter 3 presents the host organisation and the internship’s place in it. Chapter 4 formalises the problem and states the research questions; Chapter 5 describes the three approaches and Chapter 6 what they measured. Chapter 7 discusses the findings, with the ethical and sustainability aspects; Chapter 8 states the contributions and the future work.

## 2 State of the Art

This chapter fixes the task and its metric, the encodings the learned recognisers see, the data, and the two ingredients the approaches build on, and closes on the gaps that motivate them.

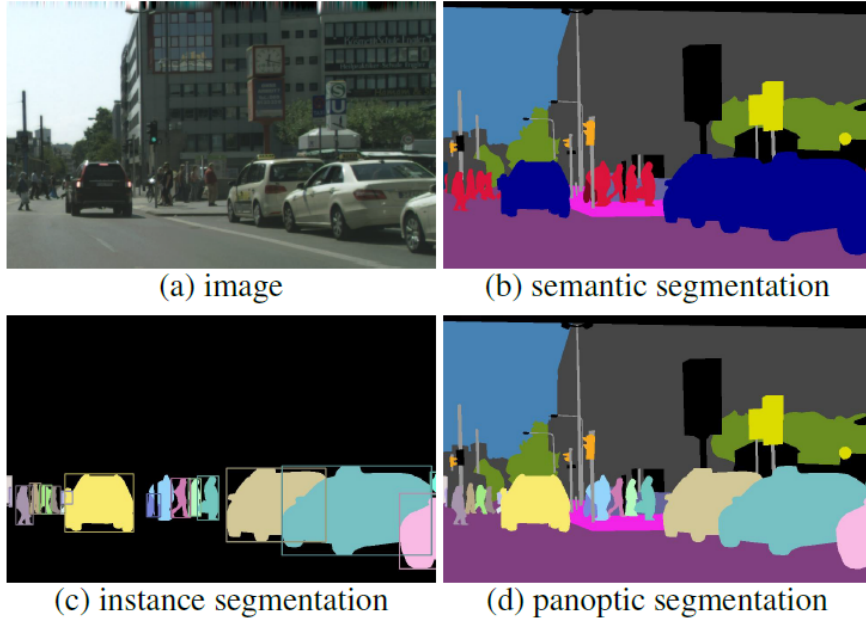
### 2.1 Machining-feature recognition

Machining-feature recognition (MFR), also called automatic feature recognition (AFR), is described by the AAGNet authors as an essential step of computer-aided process planning (CAPP), the discipline that bridges CAD and CAM by translating a design into machinable features [1, 2]. The recent literature frames MFR as *panoptic* segmentation of the face graph, assigning every face a (class, instance) pair and scoring the result with panoptic quality (PQ) [3], which FeatureFox adopted for B-reps [4]. Figure 2.1 shows what panoptic means: semantic segmentation names every pixel but does not tell two cars apart, instance segmentation separates the cars but ignores the road and the buildings, and panoptic segmentation does both. Feature recognition on a B-rep is the same task with faces in place of pixels, and stock in the role of the background. PQ scores it by matching each predicted segment to the ground-truth segment it overlaps most, counting a match only when the intersection over union (IoU) exceeds one half, which makes the matching unique; matched segments are the true positives (TP), unmatched predictions the false positives (FP) and unmatched ground-truth segments the false negatives (FN):

$$\text{PQ} = \underbrace{\frac{\sum_{(p,q) \in \text{TP}} \text{IoU}(p, q)}{|\text{TP}|}}_{\text{segmentation quality: mean IoU of matched segments}} \times \underbrace{\frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|}}_{\text{recognition quality: F1 score of the matching}}, \quad (2.1)$$

so a prediction pays both for imprecise boundaries and for the wrong number of segments, and an over-split and a wrong merge cost the same. That is what makes it a good metric for this problem. This report’s metric of record, exact instance recognition (Chapter 4), is the stricter cousin that only credits matches with an overlap of one. The instance half of the problem is the half on which both the host team’s pipeline and the public datasets are weakest [1, 4].

AFR has been studied for over four decades, and Han, Pratt and Regli [2] organise that tradition into three families: **graph matching** on the *attributed adjacency graph* (AAG) of Joshi and Chang [5], where faces are nodes, shared edges are arcs marked convex or concave, and a feature is a sub-graph matching a hand-written template; **volumetric decomposition**, which subtracts the part from its stock and decomposes the removal volume [6]; and **hint-based reasoning** [7], which generates hints for candidate features from local patterns and completes them by geometric reasoning, the family designed to handle interacting features. Rule-based recognition is still the technology of commercial software [1], and its limits are what moved the field to learning (Section 1.1). The tradition was not discarded, though; the approaches of this report re-parameterise it twice. FeatureFox, which approach 2 extends, is structurally an AAG



**Figure 2.1:** Semantic (b), instance (c) and panoptic (d) segmentation of one image, reproduced from Kirillov et al. [3], Fig. 1: panoptic segmentation gives every pixel one class and one instance. Machining-feature recognition is the same task on the faces of a B-rep, with stock as the background.

whose hand-written convex/concave arc attribute has been replaced by a learned, calibrated edge classifier [4]. The agent of approach 3 is a rule-based recogniser of a new kind: its rules are the labelling conventions written in natural language, its geometric reasoning is done through kernel queries, and it applies them face by face until the part is partitioned.

## 2.2 Representations and learned recognisers

No learned recogniser sees a B-rep. It sees an *encoding* of one, and that choice separates the families of recognisers as much as their architectures do. It also separates the three approaches of Table 1.1 from each other, so the vocabulary is fixed here once. The object being encoded is a solid  $S$  whose boundary is the union of its faces. A face  $f$  is a *trimmed parametric surface*, a smooth map

$$\sigma_f : \Omega_f \subset [0, 1]^2 \rightarrow \mathbb{R}^3, \quad n_f = \frac{\partial_u \sigma_f \times \partial_v \sigma_f}{\|\partial_u \sigma_f \times \partial_v \sigma_f\|}, \quad (2.2)$$

in which  $\Omega_f$  is the trimming domain, the part of the parameter square that actually belongs to the face, and  $n_f$  is the outward normal; an explicit topology records which faces share which edges. Every encoding below turns that exact description into a fixed-shape tensor, and each does so by throwing something away.

**Voxels.** Occupancy on a regular grid. The first deep recogniser, FeatureNet [8], classified voxelised blocks carrying one feature each, and the field left the representation behind: cost grows with the cube of the resolution, while a chamfer band or a small bore, exactly the geometry a machining feature turns on, is thinner than one voxel at the resolution that fits in memory [1].

**UV grids.** Evaluate each face on a regular  $m \times m$  grid of its  $(u, v)$  parameter square,

$$G_f[i, j] = (\sigma_f(u_i, v_j), n_f(u_i, v_j), \mathbf{1}[(u_i, v_j) \in \Omega_f]) \in \mathbb{R}^7, \quad G_f \in \mathbb{R}^{m \times m \times 7}.$$

The samples come from the kernel, so positions and normals are exact, and every face yields the same tensor shape however complicated it is; but a grid describes one face in isolation, so every cross-face relation has to be carried by a graph laid over the grids. UV-Net [9], AAGNet and my own May prototype (Section 5.1) take this route.

**Point clouds.** Sample  $N$  points on the boundary itself,  $P = \{(x_k, n_k)\}_{k=1}^N$ , and keep an index  $\iota(k)$  recording which face each point came from, so that per-point predictions can be pooled back onto faces. The whole part is a single tensor, so global context is free. Decisively for approach 1, it is also the only representation for which large pretrained encoders with released weights exist (Section 2.4), and this is what makes self-supervised transfer possible at all; self-supervision directly on B-reps exists only at a far smaller scale [10, 11]. What is lost is what the B-rep was for: topology is discarded, so face identity has to be re-attached through  $\iota$  and adjacency re-learned from positions, and a fixed budget may under-sample small faces. ASIN [12] was the first recogniser on this representation to predict face *instances*, through a similarity embedding that is then clustered; the point-cloud approach of this internship is its modern descendant.

**The face-adjacency graph with engineered attributes.** Nodes are faces and arcs are shared edges, each carrying a hand-designed vector: surface type, area and curvature bounds for a face; dihedral angle, convexity, tangency and shared-edge length for an arc. This is the oldest representation in the field, inherited directly from the AAG, and it is still the most sample-efficient: small enough to train a gradient-boosted classifier from a few hundred parts in seconds, and every attribute interpretable. Its costs are structural: the attribute vocabulary is fixed, so anything the designer did not anticipate is invisible to the model, and there is nothing to pretrain on. A family of graph networks on this representation [13–16] has saturated the semantic benchmarks, but, being semantic-only, cannot separate individual feature instances without further processing [1]. FeatureFox uses this representation, and so does approach 2, which extends it; approach 3 queries the same attributes on demand instead of computing them all in advance.

**Instance-level recognition.** Two recognisers predict instances on the face graph, and both learned approaches of this report descend from them. AAGNet [1], the reference method and the source of the MFInstSeg benchmark, attaches a UV grid to every face and edge of the attributed graph and scores every face pair for same-instance membership with a sigmoid over the dot product of two learned embeddings, trained against the ground-truth same-instance matrix; up to the encoder, that is the architecture both learned approaches reuse. FeatureFox [4], developed by my supervisor and three colleagues in the host team, replaces the learned encoder with a calibrated gradient-boosted classifier over the attributes of each shared edge, cut at a fixed threshold, and recovers instances as the connected components of the pruned graph; its claim is sample and compute efficiency, at near-parity with AAGNet on MFInstSeg. Crucially for this internship, FeatureFox also trains and reports on about 270 hand-labelled, industrially

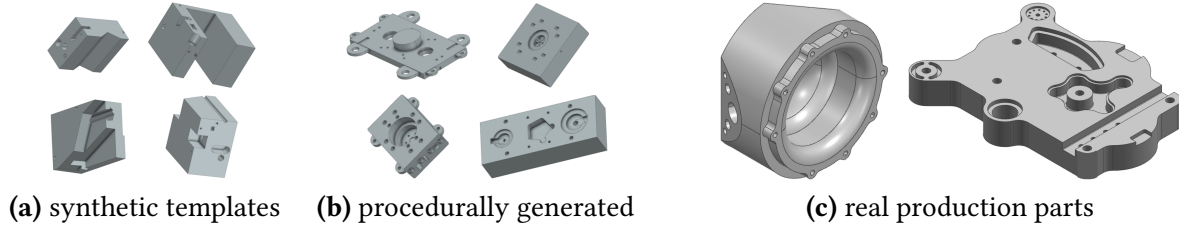
realistic Siemens NX parts, the only real-part evaluation in the instance-level literature, and it is qualitative. The paper is explicit about its boundaries, and I quote two because approach 2 inherits one and attacks the other: “real parts contain multi-operation and intersecting features whose instances may be disconnected face sets in the B-Rep yet must be treated as a single manufacturable feature – our method does not handle such cases” [4], and “a systematic cross-distribution evaluation is left for future work” [4]. Approach 2 extends this pipeline, keeping its attributes, its classifier and its decision layer, and departs from it on both points: it scores non-adjacent candidates, which is what makes the disconnected-face-set case reachable at all, and it evaluates quantitatively, cross-validated and on real parts. One of the few other quantitative real-part numbers comes from MaProNet [17], a per-face process classifier reporting around 86 % F1 on 30 industrial parts.

**From pairwise scores to instances.** Most methods that predict instances stop at the same two-step layer [1, 4, 18, 19]: score pairs, then convert the scores into a partition. The cheapest conversion, and the one AAGNet and FeatureFox use, is to threshold the pair scores and take the connected components, formalised as Equation (4.3) in Chapter 4 and solved with a union-find structure [20]. Formally the conversion is the *correlation clustering* problem [21], which is NP-hard in general, so the cheap solver is an approximation with a known failure mode. Connected components are a transitive closure: one false-positive link merges two features, one false-negative link along a thin neck splits one, and a single global threshold must trade the two off across a whole corpus. A richer solver [22], or a mask decoder that skips pair scoring [23, 24], pays off only if the residual error lies in the solver, not in the scores; Section 6.2 measures which, on real parts.

## 2.3 Data: three ways to get labelled parts

Feature-recognition data comes in three kinds (Figure 2.2, Table 2.1), and what a recogniser can learn is bounded by which of them it can be trained and measured on. **Synthetic templates** stamp a fixed vocabulary of parametric features onto a prismatic blank [1, 8, 13, 14]: every face is labelled and the corpora are large, but geometry and labels come from the same narrow generator, so a model need only re-recognise the vocabulary that made them. **Procedurally generated parts** come from programs, either hand-authored and rule-based or, as in Zero-to-CAD [25], written by a language model: richer geometry and real feature interactions, but a diversity bounded by their author and no feature labels unless the program is made to yield them. **Real parts from production** are the target: they have the complexity, the diversity and the interacting features that make the problem hard; they are the geometry a recogniser has to work on; and no labelled corpus of them is public. The FeatureFox industrial set of about 270 hand-labelled Siemens NX parts is the only one in the literature and is not released [4], and the real B-reps available at scale [26, 27] carry no instance labels. That is why so much of this internship was spent labelling, and why its hand-labelled corpora, each under its own label version, are never compared with one another.

Z2C-Seg was the internship’s attempt to close that gap. Each Zero-to-CAD part comes with



**Figure 2.2:** The three kinds of CAD data: labelled templates at scale (a, MFInstSeg [1]), procedurally generated, rule-based parts with interacting features (b), and real production parts (c) that are the target and have no public labelled corpus.

the CadQuery [28] Python script that built it, and a program is a construction history: executing it under an instrumented kernel yields a per-face instance label for 961,243 parts at no annotation cost (Section 5.2.1, Appendix A). That was the bet, and it only half paid. The parts are richer than the templates of Figure 2.2(a) but well short of the rule-based parts of (b), let alone the real parts of (c). The labels follow the modelling steps rather than a machinist’s reading of the finished part: dominated by single-face instances and far simpler than the team’s hand labels, they gave the encoder much less signal than their number suggests (Section 6.1), and the same parts proved too simple to help the real-part model. Z2C-Seg is not a corpus of real parts, and this report does not claim it is.

One limitation is shared by every labelled corpus: the labelling *convention*, which faces a chamfer, a fillet or a floor “belongs” to, is a choice and not a fact, and it bounds what any recogniser can be measured to achieve. No published corpus reports a census of its own labels; Section 6.2 reports one for the host team’s corpus (Appendix B.6), and finds the convention worth several points of exact instance recognition.

**Table 2.1:** The corpora of this report, by kind of data.

| Kind                                     | Corpus                                                                  | Parts                     | Feature labels                      | Role here                          |
|------------------------------------------|-------------------------------------------------------------------------|---------------------------|-------------------------------------|------------------------------------|
| synthetic templates                      | MFInstSeg [1]                                                           | 62,495                    | per-face class and instance         | decoder search, baselines          |
| procedurally generated, program-labelled | Zero-to-CAD [25] → Z2C-Seg (this work)                                  | 961,243 of ~1 M           | per-face instance, from the program | pretraining and probes             |
| real production, hand-labelled           | FeatureFox industrial set [4]; curated corpus, production pool, bundles | about 270 182 283; 86–254 | manual instance (and class)         | approaches 2 and 3 (Section 5.3.3) |

## 2.4 Self-supervised point-cloud encoders: Point Transformer V3, Sonata and Utonia

Approach 1 rests on three pieces of recent work, and this section explains what each is and how it is used.

**Point Transformer V3 (PTv3)** [29] is the backbone. It is a transformer over point clouds that avoids the cost of neighbourhood search by *serialising* the points along space-filling curves

and attending within windows of that ordering, scaling to hundreds of thousands of points on one graphics processing unit (GPU). Its encoder produces one feature vector per point through a hierarchy of down-sampling stages, and that per-point output is all approach 1 needs: the head of Section 5.2.2 pools it onto faces. The encoder is used frozen, so its quality is what is being measured.

**Sonata** [30] is the training scheme. It pretrains a PTv3 encoder without labels by *self-distillation*: a student encoder sees masked and augmented views of a scene and is trained to match the features of a teacher, an exponential moving average of the student, on the unmasked view. Its authors diagnosed the *geometric shortcut*, a collapse of the representation onto low-level position cues that are trivial to predict, and countered it by hiding spatial information, distilling across the hierarchy and dropping the decoder, so that the encoder itself has to carry the semantics. Approach 1 uses this paradigm, resuming pretraining from published weights; that is what “continual” pretraining means in this report.

**Utonia** [31] is the initialisation model. It trains a single Sonata-style encoder across several point-cloud domains at once, indoor and outdoor scenes and object-centric CAD among them, and releases the weights. Those weights are the starting point of the continual pretraining and, held fixed, the control encoder that every pretrained checkpoint is compared against.

None of the three was ever asked to resolve a face boundary, the granularity at which instance grouping is decided, so the question approach 1 poses is whether the Sonata objective works at the resolution this problem needs. Representation quality is measured with probes trained on frozen features [32, 33], here a small non-linear decoder, and a pretraining number means nothing unless it is a *delta* against a control encoder of identical architecture under the same probe protocol, with both probes converged before comparison (Section 5.2.3).

## 2.5 Large language models for CAD and as agents

**LLMs have entered CAD through code.** DeepCAD [34] and Text2CAD [35] generate sketch-and-extrude sequences. Text-to-CadQuery [36] and CAD-Recode [37] emit executable CadQuery code, from text or from a point cloud, and code is what lets off-the-shelf LLMs edit a model and answer questions about it. CAD-GPT [38] adds a spatial-localisation mechanism to a multimodal LLM because such models struggle with accurate 3D position and orientation. The lesson is double-edged, and both edges are used in this report: LLMs are excellent at manipulating CAD *as text*, which is what lets the labelling conventions of approach 3 be handed to a model as a document, but their geometric judgement is weakest exactly where MFR needs it, in precise position, orientation and continuity.

**Vision-language models cannot measure geometry.** Benchmarks of vision-language models (VLMs) converge on one verdict: current models see but do not perceive. BLINK [39] puts humans at 95.7 % on classic perception tasks against frontier models near 50 %, and Rahmanzadehgervi et al. [40] show frontier VLMs failing to decide whether two circles overlap. VLMs can still be useful, since images help ground the model’s prediction, but most geometric and topological analysis must be delegated to deterministic tools that return their results as text.

**Tool-using agents.** The mechanism by which a model can ask a kernel for what it cannot see is tool use. Toolformer [41] teaches a model to call external interfaces; ReAct [42] interleaves reasoning and acting so that each observation can change the next query, and software agents now follow that loop, with a small set of tools, budgets and a terminal action that commits [43]. Used as *annotators*, LLMs have matched crowd workers on text labelling at a fraction of the cost [44]; approach 3 asks whether the same holds for geometry once the model is given exact measurements instead of pixels. To my knowledge no prior work lets an LLM agent recognise machining features by querying a B-rep; the closest analogue, CAD-Assistant [45], drives FreeCAD through Python tools to edit sketches and answer questions about CAD models, but recognises no machining features; CAD-Recode and Zero-to-CAD [25, 37] only generate geometry.

## 2.6 Research gap

Three gaps run through the review, and the three approaches answer one each. **Data and label efficiency:** no public corpus is at once realistic, instance-labelled and large. Self-supervised pretraining, the standard answer to a label shortage elsewhere in 3D vision [30, 46], has been developed at the resolution of a point rather than of a face boundary, and nobody has measured the exchange rate between pretraining and hand-labelled parts (approach 1). **What limits recognition on real parts:** the scores, their conversion into a partition, the volume of labels or their quality could each bound exact instance recognition on real parts. Which one does is measurable, yet the literature never measures it, because it evaluates on corpora where the question does not bite (approach 2). **Recognition without training:** nobody has tested whether a model that reads the conventions as text and queries the kernel for exact facts can recognise instances without task-specific training (approach 3). Cutting across all three is the convention limitation of Section 2.3: the field has no consensus on one, and Chapter 7 argues that it is the binding constraint.

## 3 Company Context and Environment

### 3.1 Siemens AG and Foundational Technologies

Siemens AG is organised into three operating companies (Digital Industries, Smart Infrastructure and Mobility). This internship belongs to the first of them by subject matter. Digital Industries covers factory and process automation and, through Siemens Digital Industries Software, the industrial software portfolio in which Siemens NX sits. Siemens NX is an integrated platform for CAD, CAM and computer-aided engineering (CAE): the same solid model is designed in its modelling modules and prepared for production in its CAM module, where a process planner chooses tools, operations, set-ups and sequences. Siemens NX CAM is the product into which the work reported here is meant to land, and that destination decided most of the technical trade-offs of Chapter 5.

Alongside the operating companies, Siemens maintains a corporate research organisation, and it is there, not in a product unit, that I was hosted. The team code is FT RPD AMC DMT-DE. Read from the left: Foundational Technologies (FT) is the cross-business software and technology unit; Research & Pre-Development (RPD) is its applied-research arm, working three to five years ahead of the product roadmap; Advanced Manufacturing & Circularity (AMC) is one of the Siemens Company Core Technologies; and Digital Model Twin Germany (DMT-DE) is the German group of that field: some thirty to fifty people based in Munich-Perlach and Garching. That three-to-five-year horizon means work is not judged by a conference deadline or by a release date, but by whether a result is credible and transferable enough for a product organisation to adopt it two or three product cycles later.

### 3.2 The team and the FeatureFox project

Inside DMT-DE I worked in the AI CAD-to-CAM pipeline team. The team's own system for the CAD-to-CAM step is FeatureFox [4], released as a preprint by my supervisor and three colleagues and reviewed among the learned recognisers in Section 2.2. Its claim is sample efficiency and speed rather than raw accuracy: panoptic quality above 0.9 from roughly 250 training parts, against roughly 5,000 for the AAGNet baseline [1], and training measured in seconds. That claim matters to the team because a product model has to be lightweight and to work when the customer has only a few labelled parts.

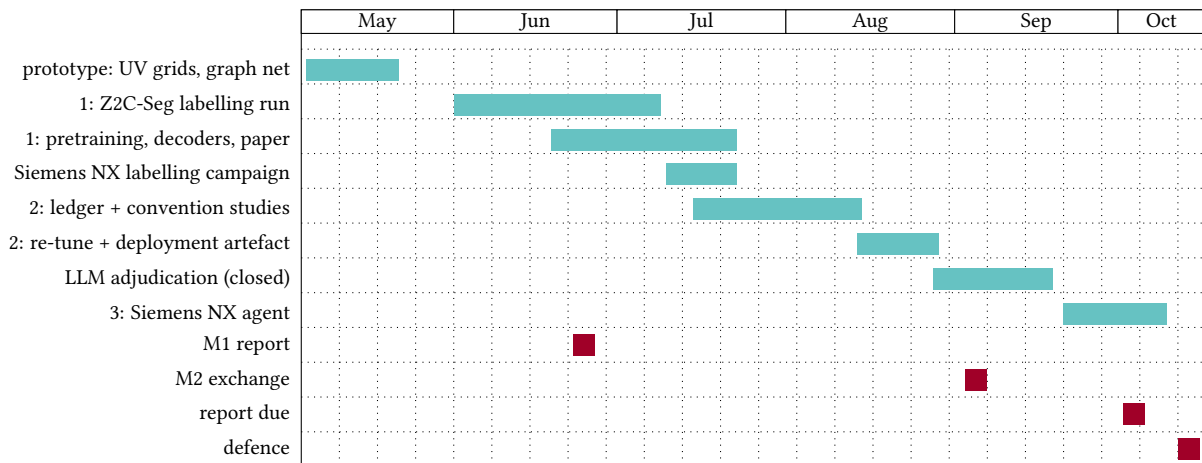
Ground truth is produced inside the product itself: the team labels parts in Siemens NX CAM. The product vision at the end of the path is a single button in Siemens NX: one click, features recognised and saved as Siemens NX feature instances, exactly as the labelling tool writes them today but automatically.

**Working arrangement.** My supervisor provided the starting tools, the Siemens NX labelling tool and his generator of templated synthetic parts, and owned the team repository. I worked on

my own branches, which he reviewed and merged into the main line; the FeatureFox extension was integrated on 11 and 14 August. I presented the Z2C-Seg labelling run to the team on 26 June. Contested labels were settled on calls with the industrial partners (Section 5.3.3). Changes of direction were argued in research status meetings.

### 3.3 Objective and positioning of the internship

The objective of the internship (Section 1.3) sits at a precise point of the team’s roadmap: between the FeatureFox recogniser, whose instance grouping is the weaker half, and the one-click Siemens NX action the product path leads to. The three approaches of Table 1.1 were explored on that objective in the order shown in Figure 3.1. The first was closed in mid-July as a documented, reusable deliverable, when my supervisor and I decided together to move the effort to the team’s real parts: the parts of Z2C-Seg are invented by a program generator, and nothing measured on them could be claimed to transfer to the partner parts. The second was carried to the deployment path in August, and the third frozen for this report in the last week of September and still being extended. An earlier LLM design that only adjudicated the classifier’s uncertain pairs was closed in September in favour of the agent.



**Figure 3.1:** The internship on the calendar. Bars are prefixed with the approach of Table 1.1 they belong to; the May prototype, the labelling campaign and the September adjudication attempt served the approaches without being approaches of their own. The red markers are the four programme milestones.

### 3.4 Resources and constraints

The material resources were modest: one Linux workstation, shared with the group, with two 48 GB NVIDIA RTX A6000 GPUs for the deep-learning work; a 36-core CPU machine for the long preprocessing jobs; my Windows laptop carrying Siemens NX 2512 for everything else; and an enterprise GitHub Copilot seat exposing GPT-5.6 Luna for the agent. Apart from the agent’s thin TypeScript extension, the stack was Python: NXOpen, the Siemens NX automation interface; CadQuery [28] on the Open CASCADE Technology kernel (OCCT) [47] for the kernel-level

labelling; PyTorch [48] with the Pointcept framework [49] for the point-cloud models; XGBoost [50] and scikit-learn [51] for the production model. Two constraints shaped the work more than the hardware: the parts are partner property and never leave the machine except through Siemens AG's enterprise endpoint; and the deployment target is a C library inside Siemens NX, so the model must be small and compilable.

### 3.5 Personal objectives

Besides the objectives of the internship itself, I set four for myself. Section 7.4 answers them at the end.

- (i) **Learn approaches I had not used before.** I came in wanting to work on something other than supervised training against a fixed dataset. Self-supervised learning, and self-distillation in particular, was the part of the field I most wanted to understand from the inside rather than from a paper.
- (ii) **Learn computer vision on 3D data.** Segmentation, 3D representations and the geometry that comes with them were new to me, and I wanted the internship to be where I picked them up.
- (iii) **Gain experience with machine learning meant for production.** A model that is only ever run in a notebook is a different object from one that has to be exported, compiled and loaded by somebody else's application, and I wanted to build the second kind.
- (iv) **See what research looks like inside a company.** I wanted to work under researchers in an industrial lab and find out how that differs from academic research: in what gets chosen, in what counts as a result, and in how the work is judged.

## 4 Problem Statement

### 4.1 Formulation

A CAD part is delivered as a boundary representation, whose faces are the trimmed parametric surfaces of Equation (2.2). The face is the atom of this problem: every quantity the models consume is a property of a face or of a pair of faces, and every label is attached to a face.

A *machining-feature instance* is the set of faces produced by one machining operation: the wall and the floor of a pocket, the cylindrical bore and its entry chamfer, the two flanks and the bottom of a slot. *Stock* is the residual blank: the faces of the raw billet that no operation touched. Feature recognition has two halves, *which* faces belong together (the instance half) and *what* the resulting feature should be called (the semantic half); only the first is the subject of this report, the second being owned by the team.

**Notation.** Throughout the report,

- $S$  is a solid and  $\mathcal{F}(S) = \{f_1, \dots, f_F\}$  the set of its faces,  $F$  their number, from a few dozen to several thousand;  $f$  and  $f'$  are two faces of it;
- $K$  is the number of instances of  $S$  and  $g_1, \dots, g_K$  the instances themselves, each a subset of  $\mathcal{F}(S)$ ;  $g$  is any one of them;
- a hat marks a prediction:  $\mathcal{P}$  is what the labeller drew and  $\hat{\mathcal{P}}$  what a method returns.

**Labelling and partition.** A labeller produces a *labelling*, a function giving every face the number of its instance; instance  $k$  is then the set of faces that carry the number  $k$ , and the instances together form the partition  $\mathcal{P}$ :

$$\ell : \mathcal{F}(S) \rightarrow \{1, \dots, K\}, \quad g_k = \{f \in \mathcal{F}(S) : \ell(f) = k\}, \quad \mathcal{P}(\ell) = \{g_1, \dots, g_K\}. \quad (4.1)$$

Every face belongs to exactly one instance. A stock face counts as an instance of its own, a *singleton*, and so does a face left unlabelled, whether by the labeller or by a method. A partial answer is therefore penalised rather than ignored, and  $\mathcal{P}(\ell)$  is a partition of all the faces of the part. The partition, not the labelling, is the real object: renumbering the instances gives a different  $\ell$  and the same  $\mathcal{P}$ , so two labellings are the same answer whenever their partitions agree, and every metric below is invariant to renumbering. The partition is an awkward target to predict directly. The number of partitions of  $F$  faces grows faster than exponentially with  $F$ , and a partition has no fixed-size encoding that a network could output.

**The pairwise route.** Approaches 1 and 2 take the same detour around the partition. For two faces, define the *same-instance indicator*

$$y_{ff'} = \mathbf{1}[\ell(f) = \ell(f')] \in \{0, 1\}, \quad (4.2)$$

which is 1 when  $f$  and  $f'$  lie in the same instance and 0 otherwise. It is a property of a *pair*, so a classifier can learn it. A model predicts  $p_{ff'} \in [0, 1]$ , its estimate of the probability that  $y_{ff'} = 1$ , for every pair in a *candidate set*  $\mathcal{E}(S)$ . Scoring all  $F(F-1)/2$  pairs would be wasteful.  $\mathcal{E}(S)$  therefore holds the pairs that share an edge and, for each face, its nearest neighbours in position, in surface type and in a descriptor space, its mirror candidates and the largest faces. Mirrored features and gap-separated coaxial stacks need the extra pairs: they form one instance without sharing an edge. A single *threshold*  $\tau$  then turns scores into a partition: the pairs whose score reaches  $\tau$  are kept as links, and the instances are read off as the connected components (CC) of the graph those links draw over the faces,

$$E_\tau = \{(f, f') \in \mathcal{E}(S) : p_{ff'} \geq \tau\}, \quad \hat{\mathcal{P}} = \text{CC}(\mathcal{F}(S), E_\tau), \quad (4.3)$$

computed with a union-find structure. A face with no kept link comes out as a singleton, which is the right answer for stock. The point-cloud encoder with its face-affinity decoder (approach 1) and FeatureFox's gradient-boosted classifier over engineered pair attributes, which approach 2 extends, differ only in how  $p_{ff'}$  is obtained. Both therefore have the same two halves, the *pair classifier* that produces the scores and the *decision layer* of Equation (4.3) that turns them into instances; the two fail differently and are measured separately.

**The agent's route.** Approach 3 takes no detour. The agent starts from a part on which every face is unassigned and declares instances one at a time, assigning faces to them and moving or releasing faces it changes its mind about, until every face belongs to exactly one instance. It never scores a pair and never needs a threshold: its answer is  $\hat{\mathcal{P}}$  itself, built directly, and it is scored exactly like approach 2.

**Why exact instance recognition is an unforgiving target.** A predicted instance is usable in CAM only if it is exactly right: the planner selects a feature and issues one operation on it, so a group missing one face machines an incomplete pocket, and a group carrying one extra face cuts into material that must stay. A ground-truth instance  $g$  therefore counts only if some predicted instance is the same set of faces,  $g \in \hat{\mathcal{P}}$ . Under the pairwise route, this requires two things at once: the kept links inside  $g$  must connect it, and no kept link may leave it,

$$(i) \ (g, E_\tau \cap (g \times g)) \text{ is connected} \quad \text{and} \quad (ii) \ E_\tau \cap (g \times (\mathcal{F}(S) \setminus g)) = \emptyset. \quad (4.4)$$

The two conditions of Equation (4.4) are strongly asymmetric. Inside the group any spanning subset of links will do; outside it, one kept link to any of the tens of surrounding faces welds a foreign face in and destroys the instance. Take a plate with one pocket: a floor, four walls and a chamfer band on the rim, six faces the convention puts in one instance, plus the six outer faces of the plate, each a singleton, so  $K = 7$ . A classifier that scores every pair inside the pocket correctly but keeps one link from the chamfer to the top face breaks condition (ii) and welds the top face in: the pocket is lost, the top-face singleton is lost with it, and one wrong decision out of a couple of dozen pairs costs two of the seven instances. There is no partial credit: on one of the team's bundles (the partner-labelled part sets of Section 5.3.3) the classifier decides 82 %

of its candidate pairs correctly while exact instance recognition on the same run is 58 %. Pair accuracy is cheap; partitions are not.

## 4.2 Evaluation

The metric of record is the same for approaches 2 and 3; approach 1 reports inst-F1. Two further families serve the pairwise route, and a cost accounting serves the agent.

- **Exact instance recognition (%)**, the FeatureFox metric of record: the share of the ground-truth instances of a part that appear, as exactly the same face set, among the predicted instances,

$$\text{ER}(S) = \frac{|\mathcal{P} \cap \hat{\mathcal{P}}|}{|\mathcal{P}|} = \frac{\#\{g \in \mathcal{P} : g \in \hat{\mathcal{P}}\}}{K}, \quad \overline{\text{ER}} = \frac{1}{|S|} \sum_{S \in \mathcal{S}} \text{ER}(S), \quad (4.5)$$

computed per part and then averaged over the parts  $S$  of a corpus  $\mathcal{S}$ . Averaging per part is deliberate: a 40-face bracket then weighs as much as a 4,500-face plate. Per-part averaging also lets parts with many singletons dilute their own errors. For the deployment path I therefore also report a *multi-face* variant, the same ratio restricted to instances of two faces or more. Exact recognition is comparable only within one corpus and one label version, and Chapter 6 never compares it across them.

- **Pair-level ranking quality**, for the pair classifiers: the area under the precision–recall curve (PR-AUC) of the scores  $p_{ff'}$  against the indicators  $y_{ff'}$  over the candidate pairs, independent of  $\tau$ . On the point-cloud side the instance-level and pair-level counterparts are *inst-F1* (one-to-one matching of predicted and true instances at a face-set IoU above 0.5) and *group-F1* (over same-instance links). PQ (Equation (2.1)) appears only where a comparison with published work on MFInstSeg is intended.
- **Cost**, for the agent: its answer is a partition, so it is scored with Equation (4.5) on the same parts as approach 2, and what is specific to it is the cost of that partition, in tool calls, wall-clock time and model credits per part. Every agent number is compared with scripted rules run through the same tools.

Exact recognition is the headline metric; wherever ranking quality is also measured, the two are reported side by side and never substituted for one another, because they disagree systematically: a feature family can raise PR-AUC while lowering exact recognition (Section 6.2). Every real-part experiment of approach 2 uses 5-fold *part-level* cross-validation (split seed 42): each part is validated exactly once by a model that never saw it, and  $\tau$  is selected on an inner 20 % hold-out of the training parts, never on the test fold. Results are the mean  $\pm$  one standard deviation (SD) over folds. A per-part *oracle*  $\tau$ , chosen with the test labels, is reported only as an upper bound on the decision layer.

## 4.3 Research questions (RQ) and hypotheses

**RQ1** Does self-supervised pretraining on a large generated corpus help face-level instance grouping, and what does it buy? *Hypothesis*: pretraining mainly buys label efficiency

rather than a higher ceiling. Answered in Section 6.1.

**RQ2** On real parts, which lever moves exact instance recognition most: labels, data volume, features, model or decision rule? *Hypothesis*: data volume. Answered in Section 6.2.

**RQ3** Can an LLM agent that reads the labelling conventions as text and queries the B-rep with exact tools recognise feature instances on real parts with no model trained for the task, and where does it fail? *Hypothesis*: yes, at a cost per part compatible with interactive use; its failures will sit on the conventions the labellers themselves apply inconsistently. Answered in Section 6.3.

**RQ4** Is the labelling *convention* itself an experimental variable: is “what counts as one feature” measurable, and does the consistency of the labels bound the accuracy any part-agnostic model can reach? *Hypothesis*: a change of convention is a change of the prediction target, so it should move the metric by as much as a modelling change does; and pairs whose label differs between two parts of identical geometry are unwinnable by construction. Answered in Section 6.2 and discussed in Chapter 7.

**Expected deliverables.** A production instance-recognition configuration with its export path; the agent as a runnable package inside Siemens NX; the Z2C-Seg corpus with its pretrained checkpoints and label-efficiency curves; and the labelled real-part corpora under a written convention. Table 7.1 breaks these into eight objectives with their outcomes.

## 5 Methodology

The three approaches attack the same target with three different sources of knowledge, and this chapter describes each with the protocol used to measure it, in the order in which they were built. The first two share the pairwise skeleton of Section 4.1 and differ only in where the pair score comes from: a self-supervised encoder, or a gradient-boosted model over engineered pair features. The third does not score pairs at all, and Section 5.4 explains what it does instead.

### 5.1 The shared skeleton

In the first fortnight of the internship I built a complete prototype: DINO-style self-supervised embeddings [52] of per-face UV grids, fed to a graph attention network (GATv2) [53] over the face-adjacency graph. It came within 3.2 PQ points of the published state of the art on the MFInstSeg benchmark [1, 4] (Appendix A.4). I then retired it, because that benchmark is templated and synthetic while the team’s target is real parts. The two learned approaches inherited its skeleton: a face representation, an encoder or feature map, a pairwise affinity  $p_{ff'}$ , a threshold  $\tau$  and connected components. Approach 1 instantiates it with point clouds and a Point Transformer V3 encoder, approach 2 with engineered pair features and gradient boosting; approach 3 replaces the whole chain by an agent that assigns faces to instances directly.

### 5.2 Approach 1: self-supervised point-cloud encoders at scale

The bet of the first approach is that the label shortage can be side-stepped rather than solved: if a large enough corpus of realistic solids existed, a point-cloud encoder pretrained on it without labels should need only a handful of labelled parts downstream, as self-supervision does everywhere else in 3D vision (Section 2.4). No such corpus existed, so I manufactured one, with instance labels to measure the transfer against.

#### 5.2.1 A corpus labelled by its own construction history

A face of a finished B-rep carries no record of the operation that produced it, but the program that generated the part does. The CQLABELER pipeline, which I built for this internship, executes each of the roughly one million CadQuery programs of Zero-to-CAD [25] under an instrumented OCCT kernel [28, 47]. Every solid-mutating operation is journalled with its inputs and result, and the construction sequence is recovered from the journal. A replay then mints one instance id per operation for the faces that operation creates; surviving faces keep their label. Deterministic post-passes turn modelling history into machining semantics, every written part is reloaded and re-matched face by face, and any solid whose history cannot be certified is rejected with a machine-readable reason. The result is Z2C-Seg: 961,243 instance-labelled solids, 96.2 % of those programs, with label, point-cloud, Siemens NX and STEP exchange-format exports on one canonical face order for cross-checking. Its labels are exact by construction, but they inherit the

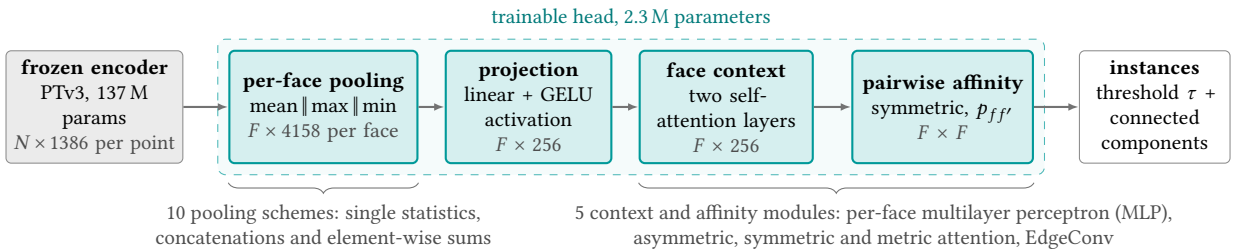
*modelling* decomposition rather than a machinist’s, and its geometry is what a program generator invents; both limits return in Chapter 7. Appendix A describes the labelling mechanism and the statistics of the corpus.

Point clouds are sampled per face with an area-proportional budget and a floor: face  $f$  receives  $b_f = \max(\text{round}(T A_f / \sum_{f'} A_{f'}), b_{\min})$  points, where  $A_f$  is its area,  $T$  the budget per part (16,384 points for most of the corpus) and  $b_{\min} = 100$  the floor, so that even a narrow chamfer face is represented. Each point carries its position, normal, face centroid, face id and instance id. The corpus is split 90/5/5 into training, validation and test parts: every point-cloud number in Chapter 6 is a validation number.

### 5.2.2 Encoder, continual pretraining and the face-affinity decoder

The encoder is Point Transformer V3 [29] initialised with the weights of Utonia [31], continually pretrained on the training split with Sonata’s self-distillation objective [30]. The adaptation to CAD geometry is minimal, so that the comparison against the initialisation stays honest: the colour channel carries the face centroid, a cue that is constant on a face and discontinuous across face boundaries; chromatic augmentations are dropped; masks keep Sonata’s room-scale defaults, which I did not ablate. The long run is ten epochs over 5.6 days on two RTX A6000 GPUs, with a checkpoint kept after every epoch ( $e_{01} \dots e_{10}$ ) and the Utonia initialisation ( $e_{00}$ ), called un-pretrained below because it has seen no Z2C-Seg data, kept as the anchor.

The decoder (Figure 5.1) pools the per-point features of each face by concatenating their mean, maximum and minimum, projects to 256 dimensions, contextualises the face embeddings with two self-attention layers, predicts a *symmetric* pairwise affinity trained with binary cross-entropy (BCE) against the same-instance indicator of Equation (4.2), and reads instances off as connected components at a threshold. It is deliberately small, about 2.3 M trainable parameters against a 137 M encoder, so that it can be trained quickly on the frozen encoder after each epoch. It was not hand-designed: ten per-face pooling schemes were crossed with five face-context modules, and the fifty cells took 20 GPU-hours in all on the frozen encoder on MFInstSeg (Appendix A.3). Two of the grid’s findings shaped the head: concatenating three statistics beats any single one, and the *presence* of cross-face context matters an order of magnitude more than its form. The winning head was fixed before any checkpoint of the long run was compared.



**Figure 5.1:** The frozen-encoder head. The encoder is never trained; only the head is, with binary cross-entropy against the same-instance indicator. The boxes show the winning cell of the fifty-cell grid (Appendix A.3), and the lower braces mark its two factors: ten per-face poolings crossed with five context and affinity modules.

### 5.2.3 Evaluation protocol: converged probes

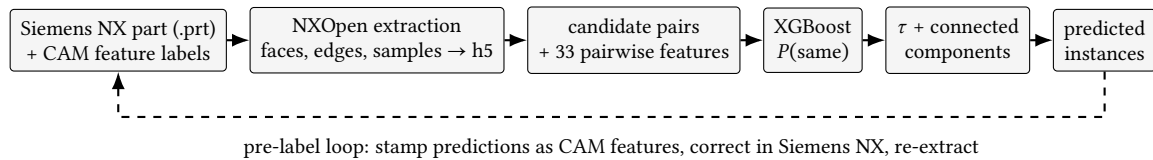
Representation quality is measured with *frozen-encoder probes*: a fresh decoder with a pinned recipe (fixed seed, fixed step count, identical data order) is trained against every checkpoint  $e_0 \dots e_{10}$ , so that a metric difference is attributable to the encoder alone. **Probes must be converged**: an under-trained probe overstates the differences between encoders; a 14-point inst-F1 gap between two checkpoints shrank to 6.5 points once the probes had converged. **Validation loss is a miscalibration signal, not a quality signal**: in a deliberate overfitting run the validation BCE doubled while validation inst-F1 kept improving, so selection is on inst-F1, never on loss.

## 5.3 Approach 2: FeatureFox instance recognition on real parts

The bet of the second approach is the opposite one: a tiny model over interpretable, engineered pair attributes, trained in seconds from a few hundred real parts, with the labels rather than the model as the thing to work on. It starts from the host team’s FeatureFox pipeline (Section 2.2) and departs from it on two points: the candidate set, which reaches beyond CAD adjacency, and the evaluation, which is quantitative, cross-validated and on real parts.

### 5.3.1 Pipeline

Parts are opened in Siemens NX through NXOpen. CAM feature groups do not survive the export to HDF5 array files, so the labels are first stamped onto the faces as attributes, a step I added; each body is then exported with its per-face and per-edge geometry (Figure 5.2). A candidate-pair generator builds the bounded set  $\mathcal{E}(S)$  of Equation (4.3), which reaches beyond CAD adjacency. This is the first departure from the published pipeline, which scores only pairs sharing an edge and therefore cannot, by construction, group a feature whose faces are separated by stock. Pairwise features, a gradient-boosted classifier and the decision layer of Equation (4.3) produce the predicted groups, which can be stamped back into the part as CAM features. That last step is the basis of the *pre-label loop*: new parts are labelled by *correcting* model predictions instead of from scratch. By my own estimate this was about five times faster, which made the data experiments of Section 6.2 affordable.



**Figure 5.2:** The FeatureFox instance-recognition pipeline and the pre-label loop.

### 5.3.2 Features, model and decision layer

Forty-three pairwise features in six families are implemented, and 33 of them form the production configuration (Table 5.1). The three families kept describe the pair on its own (geometry), the pair

in the context of the whole part (holistic) and the axis structure the two faces share (coaxiality). Only the geometry family was in the pipeline when the work started, less the adjacency flag; I added the other five. The three dropped were built, measured and rejected on evidence: cavity and transition raise pair-ranking quality and lower exact recognition (Section 6.2 and Appendix B.4), and axial interval, tried on a later gate run, raised ranking by under a point and left exact recognition flat.

**Table 5.1:** The six pairwise-feature families. “Production” marks the 33 features handed over for deployment; the other three families were ablated and dropped.

| Family         | #  | What it measures                                                                                                                                                                                                                                                                             | Production |
|----------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| geometry       | 21 | the pair on its own: normal alignment, coplanarity gap, centroid distance, adjacency and shared-edge ratio, area and perimeter ratios, surface-type match, node degrees, and, when the two faces share an edge, its type, length, smoothness, dihedral angle, concavity and continuity class | yes        |
| holistic       | 9  | the pair inside the whole part: signed centroid offsets in the part frame, summed area fraction, face-span distance, bounding-box touch counts and overlap                                                                                                                                   | yes        |
| coaxiality     | 3  | the axis structure, recovered from circular-edge frames and clustered per face: axis alignment, coaxial offset, whether the two faces share an axis                                                                                                                                          | yes        |
| cavity         | 3  | whether both faces lie in the same concave region, and what fraction of it each occupies                                                                                                                                                                                                     | no         |
| transition     | 4  | fillet and chamfer band descriptors: smooth-edge fraction and band compactness                                                                                                                                                                                                               | no         |
| axial interval | 3  | axial gap, axial overlap and radius jump on the best coaxial cluster pair                                                                                                                                                                                                                    | no         |

The classifier is an XGBoost model [50] of 200 trees at depth six, wrapped in an isotonic calibrator [54] so that the score handed to the decision layer is a probability and  $\tau$  carries the same meaning across parts. The decision layer is the global threshold and union-find of Equation (4.3), with  $\tau$  selected on a grid by maximising exact instance recognition on an inner hold-out that the test fold never sees. Seven alternatives to this layer, from correlation clustering [21] to learned per-part thresholds and a learned ranking over candidate partitions, were benchmarked on identical models (Appendix B.3).

### 5.3.3 Labelling: a three-party protocol and a measurable convention

Labels are not an input to this work but its main variable, so the labelling procedure is part of the method. Parts are labelled in Siemens NX CAM by creating one feature per instance until no face is left uncovered; the supervisor’s labelling tool provides the creation dialogue and the coverage check, and I extended it with five sub-tools for finding what is left, regrouping and auto-completing. The delivery of 222 parts from the industrial partner arrived with only 28 fully labelled; I completed and corrected the rest, and hand-labelled several hundred parts in total

across successive convention iterations. The work is split by *confidence* rather than by volume: I label what I am certain of, the supervisor labels what he is certain of in the remainder, and whatever is still contested is decided on a call with the partners who supplied the parts. The parts labelled this way form the *curated corpus* (97 parts at the start, 182 at the end): real production and demonstration parts, native to Siemens NX or imported from STEP and SolidWorks, such as moulds, housings, brackets and fixtures, drawn from the same sources as the FeatureFox industrial set. Adding 101 real parts from the team’s older foxcad corpus, which I labelled and harmonised to the same convention, gives the 283-part *production pool*. The team’s newer *bundles*, up to 254 parts labelled by the partners under their own conventions, are kept apart.

## 5.4 Approach 3: an LLM agent inside Siemens NX

### 5.4.1 Motivation and task

By the end of approach 2 the classifier was at its ceiling, and the ceiling was not the model: every gain of the last month had come from labels, and the residual errors sit where the labels themselves disagree (Section 6.2). That points at an approach in which the convention is not learned from labels but *stated*, as rules a model can read, and applied to geometry it can query. A large language model is the candidate because the conventions are already text, and because the facts that decide an instance are exact kernel queries a model can be handed as tools instead of being asked to read them off pixels, where vision-language models fail (Section 2.5). The approach is a proof of concept: can a model with no training on the task, given exact tools and the conventions, label a real part from nothing to a complete partition?

A labelled part is opened in a headless Siemens NX session and every face is scrubbed to neutral grey. The agent must group *all* faces into instances using only its tools, then call `finish_part`, which is refused while any face is still unassigned; a face machined on its own is a one-face instance. The answer is the partition  $\hat{P}$  of Section 4.1, scored with the exact-recognition metric by the same function that scores the classifier, so that agent and classifier are comparable part by part. The ground truth is read from the CAM-feature attributes of the labelled part file, for scoring only.

### 5.4.2 Architecture and tools

For each part, my Python orchestrator spawns one process of `pi`, a terminal agent client that runs GPT-5.6 Luna through the company’s enterprise Copilot seat. A thin TypeScript extension to `pi` registers the tools from the server’s manifest, injects the system prompt and enforces the budgets. Every tool is an HTTP call to a FastAPI server that drives one Siemens NX session through NXOpen and runs the tool calls one at a time; the server reads the ground truth for scoring only, and no tool can reach it. The same server also runs inside an interactive Siemens NX session, from a ribbon tab, on the part the user has open: the faces take their instance colours as the agent assigns them, and every step is written to the Information window (Figure 6.3). The tools (Table 5.2) answer the questions that decide an instance: what lies between two faces, what is coaxial with this bore, which faces are copies of this one.

**Table 5.2:** The tools the agent uses. Every tool returns text, screenshot also an image; every fact comes from the kernel.

| Tools                                                                                         | What they return                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| part_summary,<br>get_face, get_faces,<br>get_edge, neighbours,<br>relation,<br>find_similar   | the part's size, face-type and edge-class histograms and largest faces; one face's type, area, centroid, axis and radius or normal, hole-or-boss sense, coaxial faces and narrow-band hints, and every edge with its neighbour, length, interior angle (180 degrees tangent, below concave, above convex) and continuity; neighbours by edge class; coaxial, coplanar, parallel and distance relations of a pair; faces of the same type and size |
| propose, judge,<br>accept; assign,<br>unassign, instances,<br>remaining;<br>finish_part, save | the next rule group of unassigned faces with its spots; a verdict, reason and confidence on one spot; the group assigned with the verdicts applied and re-split into its pieces; by-hand assignment and release of faces, what has been declared, what is left; the end of the run, refused while any face is unassigned; a copy of the coloured part                                                                                             |
| set_view, rotate,<br>zoom, look_at,<br>screenshot                                             | view presets or azimuth and elevation, framing on faces; a render with every instance in its own colour, face numbers on request and a list of which are visible                                                                                                                                                                                                                                                                                  |

### 5.4.3 The guided walk

Left free to group faces itself, the model scored below a purely geometric grouping and varied widely between runs; the walk fixes both. Four geometric rules of my own give a first grouping: weld two faces across every concave or tangent edge; weld them across convex edges up to 200 degrees; attach a chamfer or round at the mouth of a hole to the hole; and weld the shoulder and the two bores of a counterbore. On the 78 development parts these rules alone recover 72 % of instances exactly, and their errors are concentrated: two thirds of the misses are over-merges, where one face touches several features and the rules chain them into one group. *propose* hands the agent the next rule group of unassigned faces, largest first, together with the *spots* where the rules are measurably unreliable: a *bridge*, one face whose removal splits the group into pieces, with the question of whether that face belongs with each piece; and a *flat*, a planar face tangent to outside rounds (convex rounded edges), with the question of whether it stays with them. Each spot describes the pieces and states the rate at which the labellers kept that kind of pair together. The agent answers every spot with *judge*: a verdict, a one-sentence reason and a confidence from one to five; a verdict is applied at confidence four or five; below that the labellers' majority for that kind of spot applies. *accept* then cuts the links judged apart, makes each remaining connected piece its own instance and names it; a proposal without spots is accepted as it stands, and the loop runs until no face is left (Figure 5.3). The agent commits every instance through *accept*; the proposals are deterministic, so a run is reproducible up to the verdicts.

### 5.4.4 Prompt, anti-leakage and protocol

The system prompt has five parts: the task; the walk; the two kinds of spot and what together and apart mean for each; the confidence rule; and the conventions. The conventions come from a versioned file (Appendix C.1). They are the numbered rules of the labelling guideline, rewritten for an agent that must commit rather than abstain, each with the rate at which the

```

PROPOSAL 1 (19 faces unassigned): 7 faces F1 F5 F14 F15 F16 F17 F18
held together by: 6 tangent, 5 convex, 1 concave
s1: BRIDGE F14 (cylindrical r=42.5 outer, 13 % of the part) holds 2 pieces together:
  P1: F1 (planar); reached through a concave edge (a wall meeting a floor)
  P2: F5 F15 F16 F17 F18 (2 cyl r=8 inner, 2 cyl r=21 outer, 1 cyl r=91 outer);
    F14 reaches it only through tangent edges
    is F14 the same machining feature as P1? labellers kept such pairs together 49 %
s2: is F14 the same machining feature as P2? labellers kept such pairs together 23 %
judge(s1, together, "F14 is the wall rising from F1 at a concave foot: one boss", 5)
judge(s2, apart, "the cylinders reach F14 only by tangent contacts: separate holes", 4)
accept("raised circular boss wall and base")
-> 2 instances after the cuts: #1 F5 F15 F16 F17 F18, #2 F1 F14; 7 of 19 assigned

```

**Figure 5.3:** One proposal of the guided walk on a held-out part, condensed from the tool log; the rules' group, its two bridge spots with the pieces described and the labellers' rate, the agent's two verdicts with their confidence, and the accept that applied them. Both verdicts matched the labels; the part came out with all 14 instances exact.

labellers followed it on the development parts. The view and screenshot tools are registered, but the prompt never suggests a picture, because a picture of a spot measurably made the verdicts worse (Section 6.3). Three anti-leakage rules are non-negotiable: every face is recoloured at open, so the part's ground-truth colours never reach a screenshot; no tool returns a face name, attribute, CAM feature or original colour, and the server's smoke test asserts it; and parts are never saved in place.

Parts come from the team's latest 254-part bundle, split by part; parts above 80 faces (57 of 254) are excluded to keep the runs cheap, leaving 78 development and 119 held-out parts. Design decisions were taken on 24 of the 78 development parts, with a repeat run for run-to-run variation; held-out parts are run once per frozen configuration. Three baselines are scored with the same function: the rules alone and the classifier of approach 2, from its out-of-fold predictions, on every part (the classifier on 117 of the 119 held out); the free-form agent on the 24 development parts only. Budgets are 30 tool calls plus 4 per face, 12 screenshots and 30 minutes per part. The configuration reported is the frozen one: the walk with bridge and flat spots, the confidence gate, conventions version 3.

## 6 Results and Findings

### 6.1 Approach 1: what pretraining at scale buys

**The corpus.** Of the roughly one million programs, 961,243 were labelled, verified and exported (Table A.1). A median part has 29 faces and 14 instances, and 73.8 % of the instances are single faces. That last number sets the floor of every metric on this corpus: a decoder that left every face on its own would match three quarters of the instances. As an instance has 2.1 faces on average, most of that decoder’s predictions are wrong, yet it scores about 0.5 inst-F1. The numbers below measure how well the remaining quarter is grouped.

**What pretraining buys.** Table 6.1 compares, at four label budgets, a decoder trained on the frozen Utonia encoder with one trained on the best continually pretrained checkpoint. The gain is largest where labels are scarce: a pretrained encoder with 300 labelled parts beats the un-pretrained one with 3,000, so at the smallest budget pretraining is worth at least a tenfold label multiplier, and about five points at the full budget. Figure 6.1 adds the finding that matters more than the size of the gain. For every label budget the best checkpoint is an early one, epoch 2 to 4 of ten, and the gain then declines, until at the full budget the last checkpoint is no better than the un-pretrained encoder. Longer continual pretraining is not better pretraining, a drift the pretraining papers of Section 2.4 do not test for.

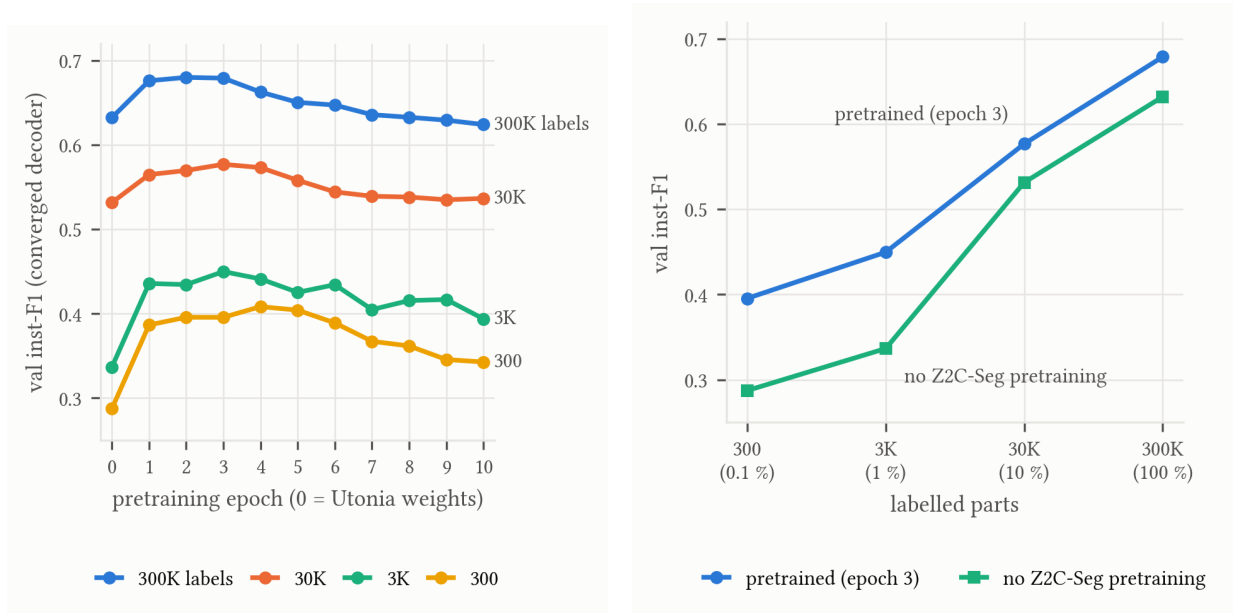
**Table 6.1:** Label efficiency on Z2C-Seg (best- $\tau$  inst-F1, 20k validation parts, single seed): Utonia initialisation  $e_{00}$  versus the best continually pretrained checkpoint. Budgets are % of a 300k subset of 865k training parts;  $\Delta$  in inst-F1 points ( $\times 100$ ).

| Labelled parts  | $e_{00}$ (no Z2C-Seg pretraining) | Best pretrained (epoch)   | $\Delta$ (points) | $\Delta$ rel |
|-----------------|-----------------------------------|---------------------------|-------------------|--------------|
| 300 (0.1 %)     | 0.287                             | <b>0.408</b> ( $e_{04}$ ) | +12.1             | +42 %        |
| 3,000 (1 %)     | 0.337                             | <b>0.450</b> ( $e_{03}$ ) | +11.3             | +34 %        |
| 30,000 (10 %)   | 0.532                             | <b>0.577</b> ( $e_{03}$ ) | +4.5              | +8.5 %       |
| 300,000 (100 %) | 0.632                             | <b>0.680</b> ( $e_{02}$ ) | +4.8              | +7.6 %       |

**Baselines.** AAGNet [1], retrained class-agnostically on the same split, reaches validation inst-F1 0.175, against 0.632 and 0.680 for my decoder on the frozen encoder without and with Z2C-Seg pretraining. AAGNet scores below the 0.5 that leaving every face alone would give: it predicts a dense link matrix, and with the positive weight it needs to learn at all (Appendix A.4) it over-links on a corpus where most instances are single faces. I did not re-tune it, so the gap measures this configuration, not graph networks as such.

### 6.2 Approach 2: which lever moves exact recognition on real parts

The three corpora of Section 5.3.3, the curated corpus, the production pool and the bundles, appear in this section and are never compared with one another (Section 4.2).



**(a)** converged decoders versus pretraining epoch, **(b)** label efficiency: epoch-3 encoder versus no Z2C-Seg pretraining

**Figure 6.1:** What continual self-supervised pretraining buys on Z2C-Seg (validation split, single seed, best decoding threshold). (a) Every label budget peaks at encoder epoch 2–4 and then declines; at full labels the last checkpoint falls below the un-pretrained baseline. (b) The gain concentrates in the low-label regime: 300 labelled parts with pretraining beat 3,000 without.

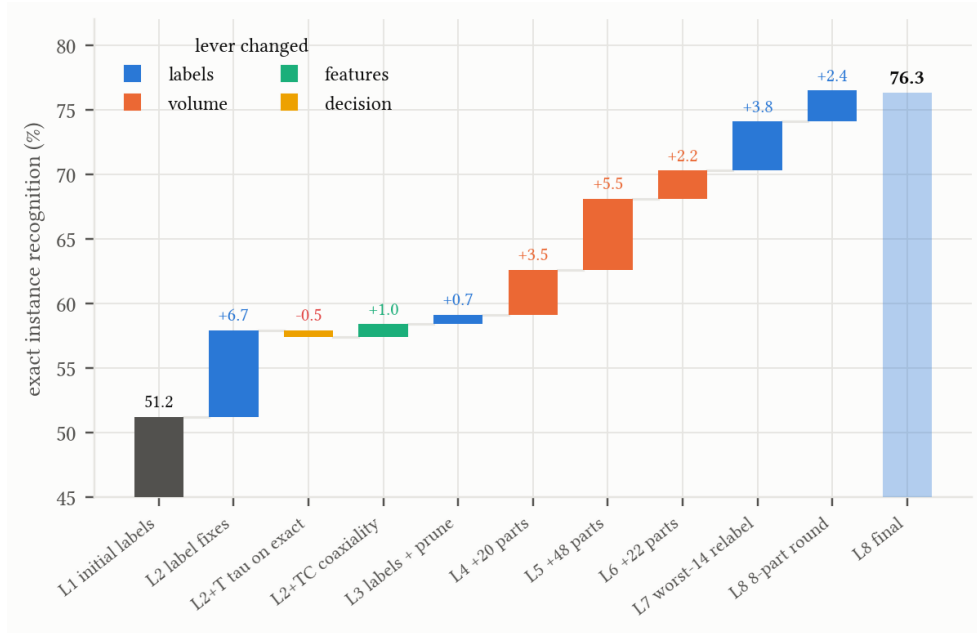
**The ledger.** Every experiment on the curated corpus changed exactly one thing, data, features, model or decision rule, and was scored against the run it descends from, so that the gain can be attributed step by step (Appendix B.1). Table 6.2 lists those steps: seven data steps, from the first label fixes to an 8-part mini-round, one feature family, one change of decision rule, monotone constraints tried on two data steps, and two feature families measured after the last data step and rejected.

**Which lever moved the number.** Figure 6.2 sorts the gain of the ledger by the lever each step changed: labels +13.6, in-domain volume +11.2, features +1.0, decision rule −0.5. Label quality is the largest lever: all four review rounds relabelled parts without adding any (Appendix B.2). Volume is second and bounded, since the sample-efficiency curves behind the data steps flatten at about 90 training parts per fold, and generous, since those steps also grew the test folds (Appendix B.1). The one feature family that helped is coaxiality, whose aggregate gain is small but whose targeted effect is not: the groups the model merged wrongly with high confidence fell by 41 %, and the worst part with a hole pattern went from 13 to 85 % exact. The monotone constraints of rows L3+M and L4+M, a restriction that forces the score up with axis alignment and down with coaxial offset, are left out of the attribution because they did not survive their own replication: a gain on one data step became a loss on the next, as spurious gains do on a corpus this small. The decision layer is not where the points are either: seven alternatives to the global threshold, benchmarked on identical scores (Appendix B.3), gained under a point at best. A per-part oracle threshold chosen with the test labels sits eight to twelve points higher, mostly

**Table 6.2:** Main ledger, curated corpus (5-fold part-level cross-validation, mean  $\pm$  SD).  $\Delta$  is against the stated baseline, which isolates one change; “Train” is training parts per fold. L2+T selects  $\tau$  on exact recognition, not pairwise F1.

| ID                             | Changed                         | vs.   | Parts | Train | Exact (%)               | $\Delta$ | PR-AUC (%)     |
|--------------------------------|---------------------------------|-------|-------|-------|-------------------------|----------|----------------|
| L1                             | initial labels                  | –     | 97    | 62    | 51.2 $\pm$ 4.1          | –        | 67.6 $\pm$ 3.0 |
| L2                             | data: label fixes               | L1    | 97    | 62    | 57.8 $\pm$ 1.3          | +6.7     | 64.9 $\pm$ 4.6 |
| L2+T                           | decision: $\tau$ tuned on exact | L2    | 97    | 62    | 57.3 $\pm$ 3.3          | –0.5     | 64.9 $\pm$ 4.6 |
| L2+TC                          | features: coaxiality            | L2+T  | 97    | 62    | 58.3 $\pm$ 3.4          | +1.0     | 65.7 $\pm$ 4.1 |
| L3                             | data: labels + prune 5 parts    | L2+TC | 92    | 58    | 58.9 $\pm$ 3.8          | +0.7     | 63.5 $\pm$ 4.9 |
| L3+M                           | model: monotone constraints     | L3    | 92    | 58    | 60.3 $\pm$ 3.6          | +1.4     | 63.5 $\pm$ 4.8 |
| L4                             | data: +20 parts                 | L3    | 112   | 71    | 62.4 $\pm$ 2.4          | +3.5     | 64.0 $\pm$ 5.5 |
| L4+M                           | model: monotone constraints     | L4    | 112   | 71    | 60.0 $\pm$ 6.1          | –2.4     | 64.5 $\pm$ 4.6 |
| L5                             | data: +48 parts                 | L4    | 160   | 102   | 68.0 $\pm$ 2.4          | +5.5     | 65.4 $\pm$ 2.4 |
| L6                             | data: +22 parts                 | L5    | 182   | 116   | 70.2 $\pm$ 3.7          | +2.2     | 65.6 $\pm$ 3.5 |
| L7                             | data: worst-14 relabel          | L6    | 182   | 116   | 74.0 $\pm$ 5.7          | +3.8     | 66.8 $\pm$ 5.0 |
| L8                             | data: 8-part mini-round         | L7    | 182   | 116   | 76.3 $\pm$ 3.2          | +2.4     | 67.0 $\pm$ 3.9 |
| F1                             | features: cavity                | L8    | 182   | 116   | 73.2 $\pm$ 7.0          | –3.1     | 69.2 $\pm$ 4.9 |
| F2                             | features: transition            | F1    | 182   | 116   | 72.1 $\pm$ 5.1          | –1.1     | 70.8 $\pm$ 5.1 |
| Cumulative L1 $\rightarrow$ L8 |                                 |       |       |       | 51.2 $\rightarrow$ 76.3 | +25.2    |                |

because it overfits them. The residual error is in the scores, not in the solver.



**Figure 6.2:** Where the +25.2 points came from: each atomic step of Table 6.2 as a waterfall, coloured by the lever it changed.

**Past the plateau, volume buys stability.** The plateau at 90 parts invites the conclusion that more data is useless, and a controlled experiment says otherwise (Appendix B.5). Two external sources, 97 simple generated parts and 102 complex real parts from the team’s older foxcad corpus, were added to the training folds only, with the curated test folds held fixed, so any gain

cannot come from an easier test population. Both lift the low-data end of the curve and leave the result on the full training set within noise, and once enough in-domain parts exist the hard real parts narrow the spread between folds. Beyond roughly two hundred training parts, count alone no longer moves the metric: additional simple parts on top of complex ones, or more parts of a kind already represented, leave it within noise. My reading is that only part families the corpus lacks would still move it; I did not test this, so further collection should target coverage and measure it.

**Conventions are a measurable variable.** By the last data step the model was being blamed for errors that are not its own. Table 6.3 summarises three studies of the labelling convention (Appendix B.6). A census gave every pair of adjacent faces a geometric signature and asked how often the labellers put the two in one instance: for coaxial adjacent pairs the answer ranges from never to more than half of the time depending on the part, and about a tenth of those pairs contradict an identical pair elsewhere, which makes them unwinnable for any model that sees only geometry. An A/B test designed by my supervisor then changed exactly one rule, whether a chamfer belongs to the feature it breaks into or is an instance of its own, on the same 20 parts labelled twice with identical folds. The separate-chamfer rule scores nine points higher with a third less fold spread, at the same pair-ranking quality, so the difference lies in what a global threshold can reproduce, not in what the model can learn. Two objections are on record: that rule also yields more and simpler groups, which flatters an exact-match metric, and both labellings started from groups that already followed it. The production pool was then harmonised programmatically to one convention, by applying the corpus-majority rule to every coaxial adjacent pair; that bought no headline points but cut the fold spread by a quarter and tightened the selected threshold into a narrow band, which is what a consistency intervention should do.

**Table 6.3:** The three convention studies. Exact recognition is mean  $\pm$  SD over five part-level folds.

| Study                                          | Parts | Result                                                                                                                                     |
|------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------|
| census of coaxial adjacent pairs               | 168   | labelled as one instance in 34 % of pairs pooled, from 0 to 55 % across parts; 10.9 % of pairs contradicted by an identical pair elsewhere |
| chamfer A/B: one rule changed, identical folds | 20    | chamfer merged into its feature $60.8 \pm 17.5$ , chamfer its own instance $70.2 \pm 11.3$ ; PR-AUC within 1.5 points                      |
| harmonisation of the pool to one convention    | 283   | 19,286 faces changed group; $79.1 \pm 3.5 \rightarrow 78.9 \pm 2.7$ ; selected $\tau$ between 0.65 and 0.75 in every fold                  |

**Production configuration.** The configuration on the deployment path is the 33-feature set trained on the harmonised production pool with a global threshold of 0.7:  $78.9 \pm 2.7$  exact on the pool, against about 76 to 77 on the curated real parts alone (Appendix B.5), and I report the two apart because a blended figure is the flattering one. On the team’s newer 134-part bundle the same recipe scores  $57.5 \pm 3.5$ : a harder corpus, labelled under other conventions, and not a regression. For the compiled export path (Appendix B.7) I dropped the calibrator, since `t12cgen` cannot compile it, and re-tuned the model in three ways: positive pairs weighted per instance,

so that one 547-face instance stops dominating the positive class; a larger booster; and a fixed threshold. On the first, 86-part bundle the re-tune moved exact recognition from 56.4 to 62.4 and multi-face exact recognition, the figure that matters to a CAM user, from 34.3 to 50.3, since a single-face instance needs no recognition.

### 6.3 Approach 3: an agent with no trained model

**From a blank part to a partition.** Table 6.4 collects the measurements. Left to itself, the language model scores below the geometric rules it was given, below the trained classifier on multi-face instances, and differently on every run. Walking the rules’ proposals and judging only the flagged spots made it reproducible (21 of 24 parts identical on a repeat run) and level with the rules, but its verdicts on the two kinds of spot tried first, flats and weak links (Appendix C.2), added nothing on the held-out parts. The bridge question is the one on which the model’s judgement adds something: on the held-out parts its applied bridge verdicts are right about four times in five (Table 6.5), while the labellers’ majority for that kind of bridge is right on 62 % of the development questions (Table C.2).

**Table 6.4:** Whole-part results: exact recognition, its multi-face variant and their harmonic mean (balanced); calls and credits per part (one credit is one cent). The development block is 24 of the 78 development parts. The classifier is scored identically on the same parts, 117 of the 119 held out; an agent row is one run.

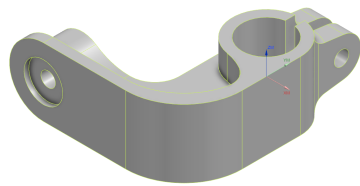
| Parts           | Configuration                        | Exact (%)   | Multi (%)   | Balanced (%) | Calls | Credits |
|-----------------|--------------------------------------|-------------|-------------|--------------|-------|---------|
| development, 24 | classifier of approach 2             | 50.9        | 58.6        | 54.5         | –     | –       |
|                 | rules alone                          | 66.9        | 54.1        | 59.8         | –     | 0       |
|                 | agent, free-form                     | 61.5        | 44.3        | 51.5         | 48    | 1.34    |
|                 | agent, walk: flats, weak links       | 69.1        | 54.6        | 61.0         | 39    | 1.04    |
|                 | agent, walk: bridges, gate           | <b>74.0</b> | <b>62.8</b> | <b>68.0</b>  | 35    | 0.90    |
| held-out, 119   | classifier of approach 2 (117 parts) | 60.6        | 64.1        | 62.3         | –     | –       |
|                 | rules alone                          | 74.0        | 61.5        | 67.2         | –     | 0       |
|                 | agent, walk: flats, weak links       | 73.9        | 59.3        | 65.8         | 37    | 0.91    |
|                 | agent, walk: bridges, gate (frozen)  | <b>76.6</b> | <b>66.6</b> | <b>71.2</b>  | 32    | 0.77    |

**Held-out parts.** On the held-out parts the frozen configuration labels every face of every part and finishes ahead of the trained classifier on the classifier’s own metrics: well ahead on exact recognition and a little ahead on multi-face instances; on the 117 parts both scored, it is better on 80, worse on 18 and equal on 19. A part takes about a minute and costs under a cent. Table 6.5 breaks the run down by part size and by verdict (by complexity class in Appendix C.3). Exact recognition falls with part size, which is what the 80-face cap anticipates. The verdict side shows why flats are gated: the model is right on well under half of its ‘apart’ verdicts on flats, against about four in five on bridges. What this proves is narrower than the margin suggests: a model with no training on the task, given exact tools and the conventions as text, labels a real part from nothing to a complete partition (Figure 6.3), and the geometric rules it walks account for most of the score.

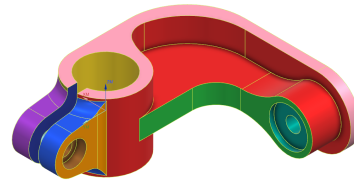
**Where it fails.** The largest remaining error is the one no rule and no verdict has resolved: whether a large flat belongs with the outside rounds it is tangent to. The labellers keep the two together in 77 % of cases, no measured geometry predicts the other quarter, and answering it perfectly would take the rules from 72 to 80 % exact on the 78 development parts; when asked, the model is right about as often as the base rate, so flat verdicts are gated to high confidence. Pictures do not help (Table C.2): a picture of a bridge spot, in any of four designs, cost five to eight points of per-question accuracy, because the model then re-describes the part in feature words and overrides the facts.

**Table 6.5:** The frozen held-out run by part size (left) and by verdict (right); a verdict is applied at confidence four or five, and *Right* means it matched the labels.

| Faces    | Parts | Exact (%) | Multi (%) | Balanced (%) | Spot   | Verdict  | Applied | Right (%) |
|----------|-------|-----------|-----------|--------------|--------|----------|---------|-----------|
| up to 20 | 36    | 79.5      | 62.8      | 70.2         | bridge | apart    | 64      | 77        |
| 21–40    | 54    | 77.8      | 69.8      | 73.6         | bridge | together | 44      | 80        |
| 41–80    | 29    | 70.8      | 65.0      | 67.8         | flat   | apart    | 67      | 40        |
|          |       |           |           |              | flat   | together | 245     | 80        |



(a) start



(b) end: 15 instances

**Figure 6.3:** The agent inside interactive Siemens NX on a 49-face bracket from the development parts, as it saw the part; 6 of 14 labelled instances exact, the largest group (red) merging two.

## 7 Critical Discussion

This chapter says which properties of the three approaches hold up and which do not, what I would do differently, what the work is worth to the team, and what it raises in terms of ethics and sustainability.

### 7.1 Strengths and limitations

**Approach 1.** The corpus is the strength: labels read off the kernel operation that created each face are exact by construction, at a scale no hand-labelled corpus reaches, and the probe protocol, a converged decoder with a pinned recipe on every checkpoint, makes the pretraining curves attributable to the encoder alone. The limitations are three. The labels follow the modelling steps of a program rather than a machinist’s reading, three quarters of the instances are single faces, and the geometry is what a program generator invents, so the label-efficiency result is a statement about Z2C-Seg and nothing else; transfer to real parts was never run. Continual pretraining drifted: utility peaked at epoch 2 to 4 and was gone by epoch 8 at the full label budget, where more than half of the 5.6 days on two GPUs bought nothing. And every point outside the 30,000-label tier rests on a single seed.

**Approach 2.** The ledger is the strength: one change per experiment, scored against its parent under one protocol, so the gain is attributed lever by lever; in the data experiments external parts went to training only. The in-domain data steps grew the test folds, so their +11.2 is generous. The limitations: the corpora are small, so a handful of parts moves the mean, and one apparent gain, the monotone constraints, reversed on the next data step; the model was never compared with a scripted rule on the curated corpus, and on the agent’s held-out parts the four rules I wrote for the agent beat it on exact recognition (74.0 against 60.6, Table 6.4), losing only on multi-face instances (61.5 against 64.1); and the model scores about 35 points higher on its training parts than on the test parts of the same folds, so it largely memorises, which is why count alone no longer moves the metric.

**Approach 3.** The strength is the measurement: the agent cannot see the labels, its baselines run through the same tools, the prompt was tuned on development parts only, and each walk of Table 6.4 was run once on the held-out parts. The limitations: parts above 80 faces were excluded, for cost; every held-out number is one run; run-to-run variation was measured on 24 development parts only; and the geometric rules account for most of the score, the model’s own judgement paying off on only one kind of decision.

**Across all three.** The labelling convention bounds every approach: about a tenth of the coaxial adjacent pairs carry opposite labels across parts of one corpus; the production pool and the newer bundle follow opposite rules on whether a bore keeps its floor or shoulder (Appendix B.6); the agent met the same wall on whether a flat joins the outside rounds tangent to it. No model or prompt can be right for both halves of an inconsistent corpus.

## 7.2 Lessons learned

- **Label quality beats volume beats features.** I began the real-part phase convinced the constraint was the number of labelled parts; the ledger says otherwise. Beyond about two hundred training parts, more parts of the kinds already present stop helping, and whether new part families would help is untested.
- **Optimise the metric that will be reported.** Two feature families raised pair ranking as exact recognition fell; a proxy rising while the metric of record falls is a warning.
- **One change per experiment, and check the instrument first.** Without the first rule the ledger's gain is an anecdote; without the second, an under-converged probe would have overstated the very differences it was meant to measure.
- **Write the convention down, with its base rates.** The largest single step of the classifier's ledger was a relabel, and the part of the agent's prompt I would keep first is the measured rate beside each rule. A labelling convention belongs in a versioned document that a pipeline, or a prompt, reads.
- **Run the cheap experiment first.** The most expensive approach produced no result on the parts the team has to machine; the classifier, which trains in seconds, produced the configuration handed over for deployment; and the agent labelled real parts for a few dollars of inference. The order in which the three were built was the wrong one for the question the team had.

## 7.3 Deployment considerations

The deployment target of approach 2 is not a Python service: the booster is exported for compilation into a C library that the FeatureFox host library will load, so that a model can be swapped in Siemens NX without rebuilding the host library. The threshold is the fragile part. A single global  $\tau$  ships, because no alternative gains more than a point on it and the decision layer has one to three points of headroom. Its selected value spreads between folds, so it must be re-selected on a held-out split whenever the data, labels or model change, never on the parts it is reported on. The agent is already an interactive tool: it runs inside Siemens NX, on the part the planner is looking at, colours the faces as it assigns them, and costs about a minute and under a cent per part. What stands between that prototype and a product is the model: a tool reply verbalises confidential geometry, so a productised agent needs a model hosted inside the company's own infrastructure.

## 7.4 Contribution and value created

**What the team keeps.** The production instance-recognition configuration and the labelled corpora behind it: the 33 features, the classifier, the threshold, the export with its frozen feature contract, the Siemens NX stamping prototype, the 182-part curated corpus and the 283-part harmonised pool. Z2C-Seg: 961,243 provenance-labelled solids with four aligned exports, ten pretrained checkpoints and their probe curves. On the Siemens NX side, five labelling sub-tools, the pre-label loop, which I estimate made labelling about five times faster, and the agent's ribbon

tab. The team also keeps the agent as a package, whose draft geometric rules alone score 74 % exact on the held-out parts.

**Objectives versus outcomes.** Table 7.1 sets each objective against what was delivered. Three fell short: the transfer of the pretrained encoder to real parts, the native build of the export path, and the dataset paper, which is parked. The four personal objectives of Section 3.5 were met. (i) Self-supervised learning and agents stopped being things I had read about; in both, the measurement turned out to be where the work is. (ii) The MSc’s machine-learning courses became working knowledge of point-cloud encoders, instance segmentation and B-reps. (iii) A result counted only once it was exportable; dropping the calibrator because `t12cgen` could not compile it is the clearest case I have of a modelling choice overruled by a deployment constraint. (iv) The changes of approach were argued with the supervisor rather than assigned. I did not expect how much of the value sits in an evaluation protocol trustworthy enough that a negative result can be acted upon.

**Table 7.1:** The objectives set at the start of the internship and during it, against outcomes at its end.

| Objective as stated                                                                      | Verdict   | Outcome                                                                                                                                                                       |
|------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Remove the annotation bottleneck with provenance-based labelling                         | ✔ done    | Z2C-Seg: 961,243 labelled solids (96.2 % of the corpus), four aligned exports. Generated parts only: the labels follow modelling steps, and the real-part bottleneck remains. |
| Test whether self-supervised pretraining at scale helps face-level grouping              | ✔ done    | +4.8 inst-F1 points at the full label budget, +12.1 points and a tenfold label multiplier at 0.1 % labels; the gain peaks at epoch 2–4 and decays.                            |
| Transfer the encoder to real parts and MFInstSeg; cluster classes                        | ✘ dropped | <b>Not achieved.</b> No Z2C-Seg-pretrained checkpoint was run outside Z2C-Seg; the class clustering was dropped with it.                                                      |
| Take ownership of instance recognition on the real-part corpus and raise it              | ✔ done    | 51.2 to 76.3 exact on the curated corpus and $78.9 \pm 2.7$ on the 283-part production pool, with a lever attribution of the gain.                                            |
| Prepare the deployable artefact and the export path                                      | 🔄 ongoing | Model, booster export, feature contract and Siemens NX stamping prototype delivered; the native build and its parity tests are not.                                           |
| Establish the labelling convention as a measurable variable                              | ✔ done    | Census, programmatic harmonisation and the chamfer A/B (+9.4 on identical folds); about 11 % of coaxial adjacent pairs unwinnable by construction.                            |
| Proof of concept: an LLM agent recognising instances inside Siemens NX, no trained model | ✔ done    | Harness, tools, guided walk and interactive Siemens NX tool built; one held-out run, 76.6 exact against 60.6 for the trained classifier on the same parts.                    |
| Publish the Z2C-Seg dataset paper                                                        | 🕒 delayed | Dataset and supplementary sections drafted; parked when the work moved to the production target.                                                                              |

## 7.5 Ethical and sustainability reflection

**Confidentiality.** The parts are real components supplied by industrial partners, who own their geometry. The agent is the one approach that sends anything outside the machine: every tool reply describes the part’s geometry in text, and it goes through Siemens AG’s enterprise endpoint. Recolouring and hidden face names protect the evaluation, not the geometry, which is why a product needs the in-house model of Section 7.3.

**Compute and energy.** Approach 1 cost 5.6 days on two GPUs for the pretraining, 20 GPU-hours for the head grid and about 6.5 hours per decoder; approach 2 trained on a CPU in seconds; the agent's runs came to about 740 credits. Energy was not metered; at 300 W per card the pretraining is bounded by about 80 kWh, more than half of it wasted.

**Automation and work.** Instance recognition proposes which faces belong to the same feature; it does not choose operations, tools, fixtures or sequence, and the process planner still makes every planning decision. The pre-label loop did not replace the labeller; it changed the job from drawing groups to correcting them, and the agent changes it again, to checking an answer. The risk, not measured here, is that a reviewer accepts a plausible but wrong grouping they would never have drawn unaided, so a pre-label loop earns its place only when the human keeps the final say and the corrections are measured.

**An LLM in the loop.** Delegating a decision to a model that cannot justify itself in terms a CAM engineer can audit differs from delegating it to a classifier. The design answers that concern in two ways: the agent can only colour faces and name instances, which the planner accepts or corrects; and every run is persisted and replayable, with a verdict and reason at every flagged spot. Two concerns remain: the model is confident where the labellers themselves split, so its confidence is no signal of correctness where it matters; and a commercial endpoint's pricing, availability and deprecation schedule are a poor foundation for a CAD feature with a decade-long support life.

## 8 Conclusion and Future Work

The objective was the instance half of machining-feature recognition: deciding which faces of a part belong to the same feature, so that Siemens NX CAM can treat them as one machinable entity. Three approaches were built and measured on it in five months, each taking the knowledge from a different place: a million generated parts, a few hundred hand-labelled real ones, or the labelling conventions as text and the part queried on demand.

### 8.1 Contributions

- **Z2C-Seg and what pretraining on it buys.** A corpus of 961,243 solids labelled by their own construction history, and the measurement that continual self-supervised pretraining on it is worth a tenfold label multiplier when labels are scarce, with an optimum at epoch 2 to 4 that later epochs lose.
- **Instance recognition on real parts, attributed lever by lever.** A ledger of one-change experiments that took exact recognition from 51 to 76 % on the curated corpus, with the attribution (labels first, volume second, features a distant third) that rejected the hypothesis I started with; the configuration and corpora on the deployment path; and the pre-label loop that made the labelling affordable.
- **The labelling convention as a measurable variable.** A census that puts a tenth of the coaxial pairs beyond any model's reach, an A/B test, designed by my supervisor and run on identical folds, in which one labelling rule moved the metric by nine points, and a harmonisation that tightened the deployed threshold.
- **An LLM agent that labels a part inside Siemens NX with no trained model.** A tool server owning a Siemens NX session, exact geometric tools, a guided walk that turns four geometric rules into proposals and asks the model only where the rules are unreliable, and the conventions as a versioned prompt. On 119 held-out parts it labels every face and finishes above the trained classifier on exact recognition, most of the margin coming from the rules it walks, at about a minute and under a cent per part.

### 8.2 Achievement of objectives

Table 7.1 records every objective against its outcome: five were met, one is unfinished (the native half of the export path), one was not achieved (transfer of the pretrained encoder to real parts) and one was postponed (the dataset paper). The research questions are answered as follows. RQ1: pretraining buys mainly label efficiency, a tenfold multiplier at the smallest label budget against about five points at the full budget, and only within Z2C-Seg, since transfer was never measured. RQ2: label quality, not data volume, moves exact recognition most on real parts, and richer features raise pair ranking but lower exact recognition. RQ3: an agent with exact tools and the conventions as text labels a real part end to end and finishes above the

trained classifier on exact recognition, and its failures sit where predicted, on the conventions the labellers apply inconsistently. RQ4: the convention is an experimental variable. One rule moved the metric by nine points on twenty parts, more than any modelling change, though the two objections of Section 6.2 make this an upper bound. About a tenth of the pairs are unwinnable by construction.

### 8.3 Future work

Recommendations to FT RPD AMC DMT-DE, in the order in which I judge they return most per unit of effort.

- **Settle the labelling conventions corpus-wide.** Write the chamfer rule into the team's labelling guideline once an A/B from blank labels confirms it, re-run the harmonisation whenever a bundle is added, and relabel the parts whose labels contradict the corpus majority. This raises the ceiling of every approach and costs less than any modelling change on the table.
- **Extend the agent to larger parts and repeat its runs.** Lift the 80-face cap, and use the determinism of the walk's proposals to vote over repeated verdicts. The open question is the flat tangent to outside rounds, which neither geometry nor the model answers yet.
- **Put the interactive tool on an in-house model.** Confidentiality requires it; escalating a failed part to a stronger model and accepting only what two runs agree on are the reliability levers the design leaves open.
- **Measure Z2C-Seg transfer, and finish the export path.** Fine-tune the epoch 2 to 4 checkpoints on the real-part corpus and score them with the same metric, which is also the missing section of the dataset paper; then build the native half of the export with golden vectors and a C++/Python agreement test.

### 8.4 Conclusion

The ranking the three approaches produced is not the one I expected in May. Six days of pretraining on a million generated parts produced a real effect, but on a synthetic corpus. A classifier that trains in seconds produced the number the team will ship. An agent that trains nothing labels parts it has never seen at least as well, and most of that score is the geometric rules it walks. Across the three, the lever that moved the metric furthest was neither the model nor the amount of data but the consistency of the labels, and behind the labels the convention that decides what a label means. That finding, and the protocol that made it visible, are what I would carry to the next problem: one change per experiment, the metric the CAM user experiences, baselines through the same harness, negative results kept.

The internship also settled part of my own path. I want to work where a model has to survive deployment, in someone else's application and on someone else's data, rather than on a benchmark. It convinced me that I add most value on the evaluation side: deciding what to measure, and making a negative result trustworthy. What it left open is whether I do that best in an industrial research group or in a product team, and my next position is the test of it.

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# A Approach 1: the Z2C-Seg corpus and the pretraining evidence

This appendix holds the material behind Sections 5.2.1, 5.2.2 and 6.1: the labelling mechanism and what it produced, the head grid that fixed the decoder, and the baselines and settings behind the pretraining curves.

## A.1 Supervision from construction history

Section 5.2.1 states the mechanism in one paragraph; Figure A.1 shows it. Every solid-mutating operation of a Zero-to-CAD program [25] is journalled with the live OCCT builder [47], the spine  $S_0 \rightarrow S_1 \rightarrow \dots \rightarrow S_n$  is walked back from the final solid, and a prior-wins replay mints one instance id per operation for the faces it creates: a face already explained by an earlier operation keeps its label, and each new operation claims only the faces nothing earlier accounts for. Four deterministic post-passes then turn modelling history into machining semantics. Profile unpacking gives each run of short edges in an extruded or revolved base profile its own instance, since these are features drawn into the sketch. Restocking returns an added body to stock when it holds at least 30 % of the part's face area, because material cannot be machined onto a part; cuts and blends are never restocked. Absorption hands stock faces enclosed by a feature to that feature, and a connected-component split breaks an instance whose faces do not touch into one instance per component. Every written part is reloaded and re-matched face by face before it is accepted.

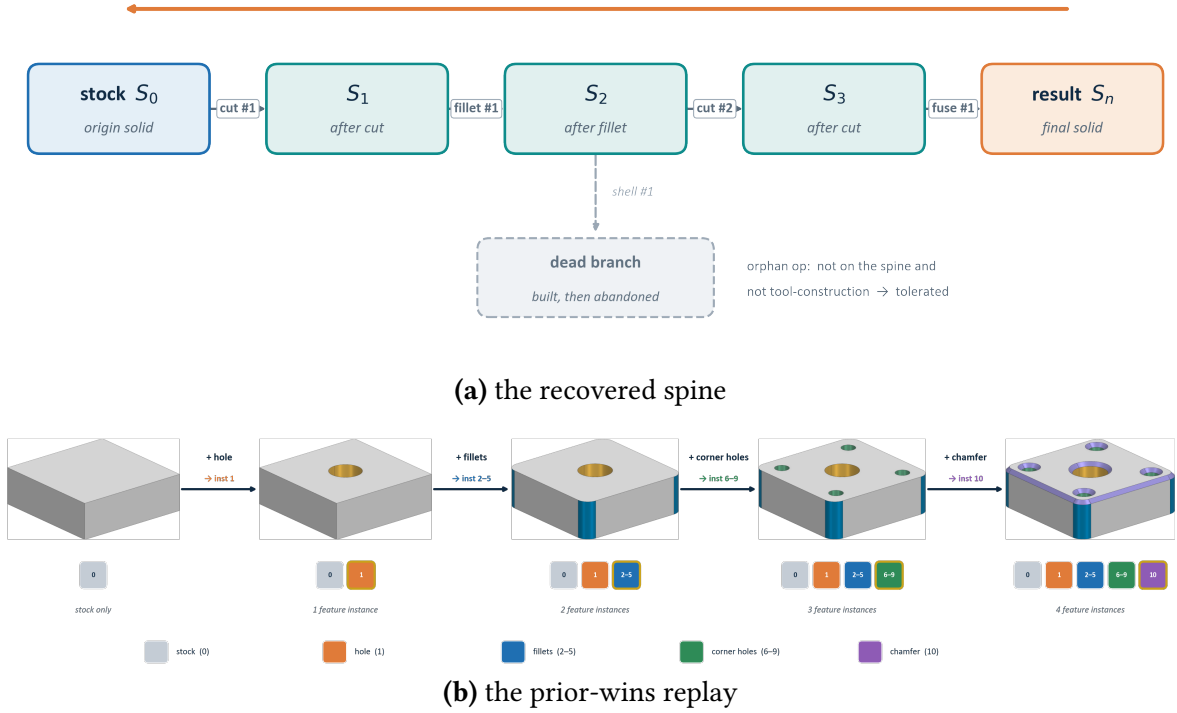
Figure A.2 shows what the labels look like on the parts, and is the picture behind the remark of Section 2.3 that they follow the modelling steps rather than a machinist's reading.

## A.2 What came out

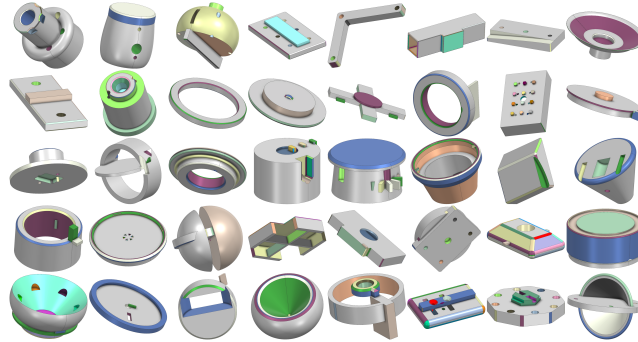
Table A.1 collects the statistics quoted in Sections 5.2.1 and 6.1; the one that governs every downstream number is the single-face share: a decoder that leaves every face on its own already matches about three quarters of the instances, which sets the floor of every metric (Section 6.1).

## A.3 The decoder head grid

The head of Figure 5.1 was fixed by the grid announced in Section 5.2.2: ten per-face pooling schemes crossed with five context and affinity modules. I trained the fifty cells on MFInstSeg under identical conditions, with the Utonia encoder frozen (cached features, seed 42, ten epochs, batch 64, one GPU, 20.2 GPU-hours in all), and selected them on validation panoptic quality [3]. Figure A.3 carries the numbers; the winning cell was also test-evaluated, at PQ 0.8844, and the ten best cells lie within 0.008 PQ of it, so the choice inside the winning family is not



**Figure A.1:** Provenance labelling in CQLABELER. (a) The spine recovered from the journal by walking back (orange arrow) from the result  $S_n$  to the stock  $S_0$ ; an operation built and then abandoned (the dashed shell) is a dead branch, tolerated but never labelled. (b) The prior-wins replay on a worked example, read left to right. Each operation mints the instance ids of the faces it creates (hole 1, fillets 2–5, corner holes 6–9, chamfer 10), and a face keeps its id once labelled. Grey faces are stock.



**Figure A.2:** Forty sample parts of Z2C-Seg coloured by feature instance, grey being stock: most coloured instances are one face each, an edge blend or a lone cut, and the geometry is what a program generator invents, richer than a templated benchmark and simpler than the real parts of approach 2.

**Table A.1:** Z2C-Seg in numbers. The operation-kind mix was measured only on the retry shard: the 111,656 parts labelled by the second run, which took the first-pass rejects with longer time limits plus the validation and test splits, and the only parts whose program metadata was kept. These parts are harder than average (median 39 faces against 29). Every other row is over the whole corpus.

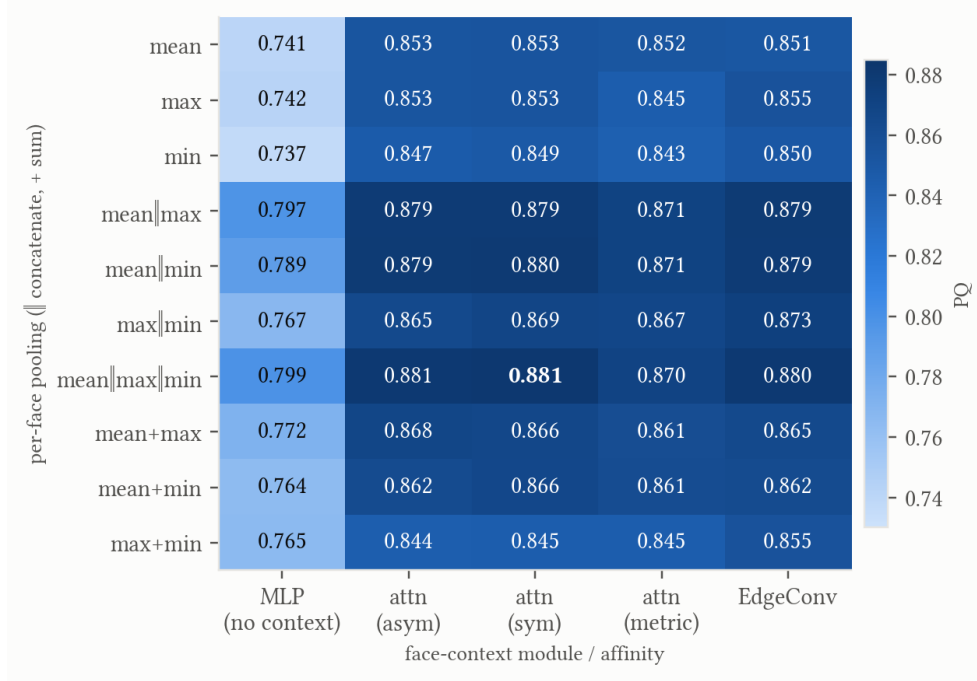
| Quantity                        | Value                                                                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Programs executed               | 999,633                                                                                                                                         |
| Labelled, verified and exported | 961,243 (96.2 %)                                                                                                                                |
| Failure reasons                 | kernel time-outs or crashes 2.1 %; round-trip order mismatch 0.6 %; script errors 0.2 %; other kernel rejections 0.1 %; Siemens NX export 0.8 % |
| Faces per part                  | median 29, 99th percentile 256                                                                                                                  |
| Instances per part              | median 14, 99th percentile 138                                                                                                                  |
| Stock                           | 23 % of faces                                                                                                                                   |
| Instance size                   | 73.8 % of instances are single-face; mean 2.1 faces per instance                                                                                |
| Operation kinds (retry shard)   | edge blends 58 %; subtractive booleans 21 %; additive booleans 13 %; shells 4 %; unpacked profile features 3 %; drafted cuts 1 %                |
| Points per part                 | about 17,000 (median)                                                                                                                           |
| Split (seed 42)                 | 865,118 train / 48,062 validation / 48,063 test; the test split was never loaded                                                                |
| Scale                           | about 15 times MFInstSeg (62,495 parts) and 27 times the Fusion 360 segmentation set (35,680)                                                   |
| Provenance                      | programs from Zero-to-CAD [25], released under Apache-2.0; labels and exports produced at Siemens AG, not yet released                          |

consequential. Three findings come out of the grid. First, at equal pooling, removing cross-face context (the MLP column, a per-face multilayer perceptron) costs 0.08 PQ, 0.7989 against 0.8814. Second, concatenation beats element-wise summation at equal inputs: mean||max reaches 0.8789 against 0.8663 for mean+max with the same decoder, and the three-statistic concatenation beats every single statistic. Third, the choice among context modules barely matters: the column means of asymmetric attention, symmetric attention and EdgeConv lie within 0.002 of each other, and metric attention, the fifth module, is about 0.006 lower (0.858).

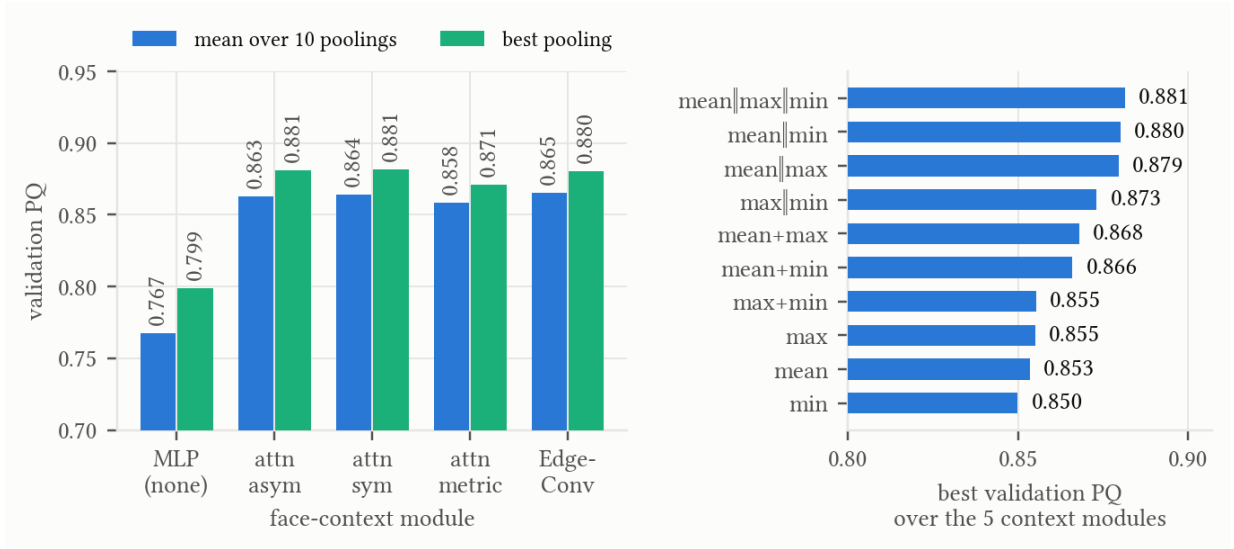
## A.4 Baselines and settings

The baseline of Section 6.1 (Table A.2) is AAGNet [1] retrained class-agnostically on the same split. I removed its bottom-face and semantic heads, re-implemented its graph extractor on OCP (the Python bindings of OCCT that CadQuery uses), and put a positive weight of 12 on the affinity loss, without which plain BCE collapses to predicting no link. Three qualifications keep the comparison honest. The run on the exact split was stopped at 40 epochs while still rising. It dropped the 107 training parts above 1,500 faces. Its test step crashed, so the comparison is on the validation set only, where the label budget favours the baseline (all 865k labelled parts against the decoder’s 300k) and the parameter budget favours the decoder (a 137M encoder against AAGNet’s 0.40M).

Table A.3 places the three systems measured on MFInstSeg during the internship beside the published numbers; it is the evidence behind Section 5.1, where the May prototype is said to have come within 3.2 PQ points of the state of the art (validation PQ 0.942 against AAGNet’s published 0.974) within a fortnight. Only the June fine-tune scored higher, by 0.006



(a) best validation PQ of every cell



(b) the two factors read off the grid

**Figure A.3:** The fifty-cell head grid on MFInstSeg, frozen Utonia encoder. (a) Rows are pooling schemes (|| concatenates the statistics, + sums them), columns the context and affinity modules; the winner, mean||max||min with symmetric-affinity attention, scores 0.8814. (b) Mean and best PQ of each context module over the ten poolings (left) and best PQ of each pooling over the five modules (right).

**Table A.2:** Supervised graph baseline versus the frozen point-cloud decoder on Z2C-Seg (AAGNet retrained class-agnostically; inst-F1 is one-to-one instance matching at face-set IoU > 0.5). The first row is the floor estimated from Table A.1; the last three rows share the 20k validation parts.

| System                                       | Split          | Training budget                     | Metric      | inst-F1       |
|----------------------------------------------|----------------|-------------------------------------|-------------|---------------|
| all faces left alone (no link)               | whole corpus   | none                                | estimate    | $\approx 0.5$ |
| AAGNet [1]                                   | 100k sample    | 100 epochs ( $\approx 7$ h)         | test        | 0.184         |
| AAGNet [1]                                   | exact 865k/20k | 40 epochs ( $\approx 24$ h), rising | validation* | 0.175         |
| decoder on $e_{00}$ (no Z2C-Seg pretraining) | exact 865k/20k | 300k labelled parts                 | validation  | 0.632         |
| decoder on $e_{02}$ (best pretrained)        | exact 865k/20k | 300k labelled parts                 | validation  | <b>0.680</b>  |

\*the post-training test step crashed with an `AssertionError`, so no test number exists for the exact split.

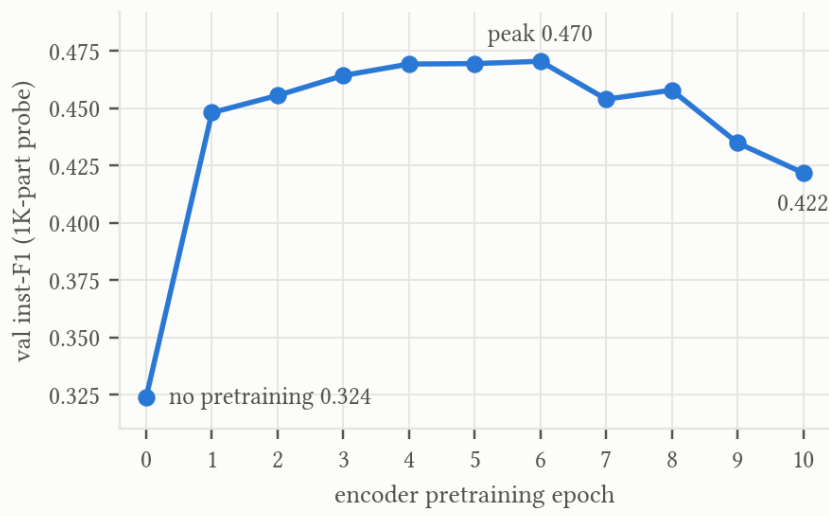
on a 2,000-part validation subset, and neither was ever test-evaluated.

**Table A.3:** Panoptic quality on MFInstSeg: the three systems of the internship against the two published ones. The May prototype (13 May 2026) and the June fine-tune (12–13 June 2026) were selected on validation and never test-evaluated.

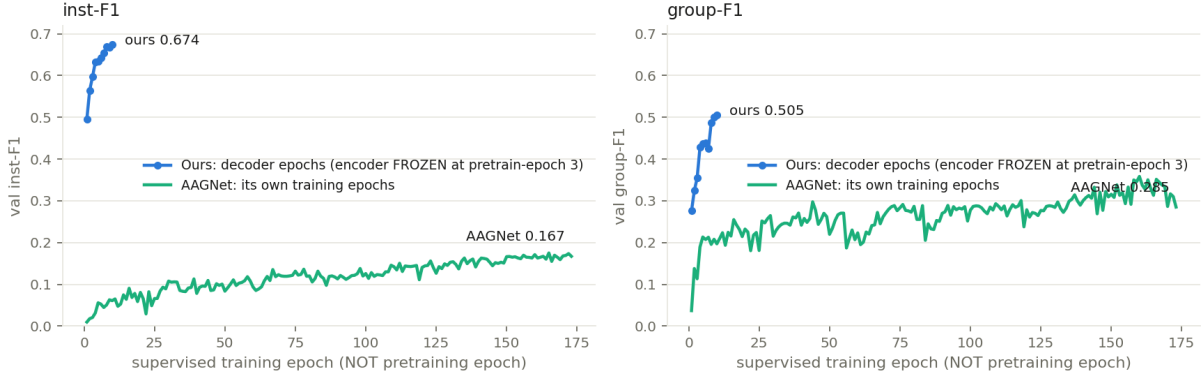
| System                                                                                                                                                         | val PQ | test PQ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| May prototype: DINO-style self-supervision [52] on $10 \times 10$ UV face grids, four GATv2 layers [53], pairwise affinity head; about 42,000 unlabelled parts | 0.942  | –       |
| June end-to-end fine-tune of the Utonia encoder with a panoptic head, 2,000-part validation subset                                                             | 0.948  | –       |
| Frozen-encoder head, the winning cell of Figure A.3                                                                                                            | 0.881  | 0.884   |
| AAGNet, as published [1]                                                                                                                                       |        | 0.974   |
| FeatureFox, as published [4]                                                                                                                                   |        | 0.969   |

Figure A.4 shows the first probe rule of Section 5.2.3 at work (a) and sets the 300k-part decoder’s training curve beside AAGNet’s (b).

Table A.4 lists the settings behind Table 6.1 and Figure 6.1; the pretraining follows Sonata [30] with the learning rate rescaled linearly to the batch that fits two 48 GB cards.



(a) the starved 1k-part probe



(b) decoder training curves, ours against AAGNet

**Figure A.4:** Two curves that need their caveats stated with them. (a) The starved 1k-part probe overstates the pretraining dividend (+14.6 points at its epoch-6 peak: 0.470 against the anchor 0.324) and misplaces the peak relative to the converged decoders of Figure 6.1: an under-trained probe exaggerates the difference between encoders. (b) The 300k-part decoder on the frozen  $e_{03}$  encoder (0.674, not the best  $e_{02}$  decoder of Table A.2 at 0.680) against AAGNet’s 100k-sample run of the same table, continued until it stopped at epoch 172, on different validation sets: the shapes are what to compare. The decoder is near its plateau within ten epochs, still rising slightly at the last; the graph baseline takes about a hundred to level off near 0.17.

**Table A.4:** Settings of the continual pretraining, the probes and the decoders of approach 1.

| Setting                                                                                                            | Value                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Continual pretraining: Sonata self-distillation of a PTV3 encoder from the Utonia weights, 137 M parameters</i> |                                                                                                                                                                                                                                                                                                                                     |
| Hardware, batch                                                                                                    | two RTX A6000 GPUs, batch 32                                                                                                                                                                                                                                                                                                        |
| Optimiser                                                                                                          | AdamW, base learning rate $1.33 \times 10^{-3}$ , layer-wise decay 0.9, OneCycle schedule, gradient clipping at 3.0                                                                                                                                                                                                                 |
| Precision, grid                                                                                                    | bf16 mixed precision; grid size 0.02 (0.01 is memory-infeasible at this batch on two 48 GB cards)                                                                                                                                                                                                                                   |
| Length                                                                                                             | ten epochs of 27,034 iterations over the 865,118-part training split, 3 to 9 July 2026 (5.6 days); one checkpoint per epoch, $e_{01} \dots e_{10}$ , plus the initialisation $e_{00}$                                                                                                                                               |
| Output                                                                                                             | 1,386 features per point; about 740 GB of feature caches                                                                                                                                                                                                                                                                            |
| <i>Probes and decoders: frozen encoder, the head of Figure 5.1, about 2.3 M parameters</i>                         |                                                                                                                                                                                                                                                                                                                                     |
| Long-run probe                                                                                                     | 1,000 training parts, 70 epochs, batch 16 on each of the two GPUs, scored on 1,000 of the 48,062 validation parts                                                                                                                                                                                                                   |
| Full-capacity decoder                                                                                              | 300,000 training and 20,000 validation parts, ten epochs, batch 256, learning rate $3 \times 10^{-3}$ , about 6.5 hours per run                                                                                                                                                                                                     |
| Label-efficiency tiers                                                                                             | 300 / 3,000 / 30,000 / 300,000 parts for 200 / 60 / 20 / 10 epochs, so that every tier is at or near convergence (the 300k decoders were still rising slightly at epoch 10); single seed, a seed replicate for the 30k tier only, on the 7 encoders (of 11) whose caches survived (mean $ \Delta $ 0.5 inst-F1 points, maximum 1.2) |
| Selection                                                                                                          | on validation inst-F1 at the best decoding threshold, never on validation loss                                                                                                                                                                                                                                                      |

## B Approach 2: the evidence behind the ledger

Every table here extends one passage of Section 6.2: the protocol and the curves behind Table 6.2, the alternatives to the decision layer of Section 5.3.2, the feature families of Table 5.1, the data experiments behind the volume claim, the three convention studies of Table 6.3, and the settings and export re-tune of the production configuration.

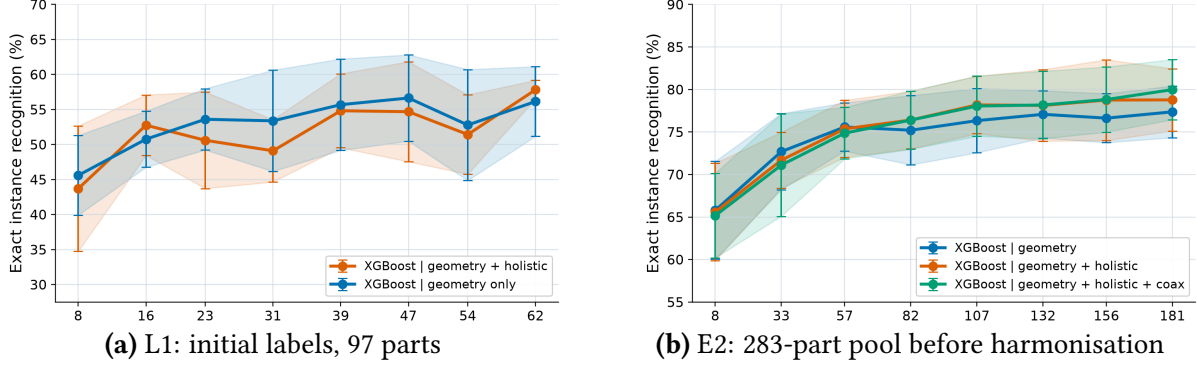
### B.1 Protocol

Every experiment of Section 6.2 runs five-fold part-level cross-validation with split seed 42, so that no part contributes faces to both sides of a fold; an inner 20 % hold-out of each fold’s training parts selects the threshold  $\tau$ , and exact instance recognition and edge PR-AUC are reported as mean  $\pm$  one standard deviation over the folds. Seven reading rules apply to every table of this appendix and to Table 6.2:

1. **A naming trap.** The experiment log and Table B.2 call the feature set of Table 6.2 geometry+holistic, but from L2+TC on it also includes the three coaxiality features. The ablation of Table B.3 redefines its base sets to exclude them, so its geometry+holistic (74.6) is not the ledger’s L8 (76.3).
2. **Data steps change the population.** The training set per fold moves 62  $\rightarrow$  58 (L3)  $\rightarrow$  71 (L4)  $\rightarrow$  102 (L5)  $\rightarrow$  116 (L6); those rows are comparable in protocol, not on identical parts.
3. **Deltas are against the stated parent and computed from unrounded fold means.** L4 is measured against L3, not L3+M, so the side rows L3+M and L4+M sit outside the chain. A delta can differ by 0.1 from the difference of the rounded columns, and for the same reason the chain of deltas, like the four levers of Figure 6.2, sums to 25.3 against the cumulative +25.2 and the rounded endpoints’ 25.1.
4. **PR-AUC has two eras.** Run F3 of 3 August, after row F2 of the ledger, corrected the metric: single-class parts are excluded from the ranking aggregate instead of counting as zero. PR-AUC values measured after it sit about two points higher by construction: Table 6.2, Table B.1, Table B.3 and Figure B.2 predate it, and Table B.7 postdates it.
5. **Corpora and label versions are never comparable**, and no table below mixes them.
6. **Names used below.** L1–L8, F1 and F2 are the rows of Table 6.2. L9 is the ledger’s configuration retrained on the 283-part production pool, the 182 curated parts plus 101 parts of the team’s older foxcad corpus, before and after its harmonisation to one convention (Appendix B.6). F3 is not a feature family but the run of rule 4. D1 and E1 add 97 generated and 102 real parts to training only (Appendix B.5), and E2 is the sample-efficiency sweep on the 283-part pool before harmonisation. Label versions follow the ledger: v8 after the L7 relabel, v9 after the L8 mini-round, v11 the 283-part pool before harmonisation and v12 after it. The bundles v0.0.1 and v0.0.2 are the team’s newer part sets of 86 and 134 parts.
7. **Re-runs differ by up to two points.** The ablation of Table B.3 is its own sweep, and Tables B.2 and B.4 re-score fixed models in a separate grouping harness, so one configu-

ration reads slightly differently from table to table: L3+M is 60.3 in Table 6.2 and 58.3 in Table B.2, and the production set is 76.3 at L8, 76.2 in Table B.3, 75.9 in Table B.4 and 75.4 in round 3 of Table B.2. These are separate runs of one protocol, not different models.

Figure B.1 shows the first and the last of the sample-efficiency sweeps behind the data steps of Table 6.2; their flattening at about ninety training parts is the plateau quoted in Section 6.2.



**Figure B.1:** Sample-efficiency sweeps at the start and the end of the ledger, exact recognition against training parts per fold (8 to 62 at L1, 8 to 181 at E2), bands  $\pm 1$  SD across folds. At L1 the two feature sets are noisy and cross; on the final pool the three nested sets rise smoothly, flatten from about ninety training parts and stay inside each other's bands. The E2 sweep predates the harmonisation and the F3 correction, so its end point of about 80 is not the production figure of Table B.9.

## B.2 Geometry-only re-run

The ledger re-run with the nine holistic features removed rises by +26.6 against the headline set's +25.2 (Table B.1). The arc of Table 6.2 therefore comes from the labelling and data work, not from the part-level context. That context buys little: +4.7 at L1 and +3.2 at L8, one to two points on most rows between, and nothing on L2+T, L3 and L4+M, where the geometry-only model is ahead. The arc survives without the holistic features; only the end points lose a few points. Every data step gains on both ledgers; the rows where they disagree in sign are the decision, feature and model steps. L2+T, the change of threshold objective, is worth  $-0.5$  on the headline set and  $+4.4$  on the weaker model, with the fold spread shrinking from  $\pm 5.0$  to  $\pm 1.2$ : the first sign that threshold selection dominates, which Appendix B.3 confirms. Coaxiality (L2+TC) is worth  $+1.0$  against  $-3.9$ , measured from that inflated L2+T baseline, and the monotone constraints flip as they did across data steps (L3+M  $+1.4$  against  $-2.2$ , L4+M  $-2.4$  against  $+0.1$ ).

## B.3 Decision-layer alternatives

A per-part oracle threshold scores eight points above the global threshold of Section 5.3.2 with the production feature set, and twelve with the earlier models of rounds 1–2, so the obvious hypothesis was that per-part adaptivity is recoverable; seven alternatives and two oracles were benchmarked against the global threshold on identical models and probabilities, only the grouping rule differing (Table B.2).

**Table B.1:** The ledger re-run on the geometry-only feature set, with folds, seed, threshold selection and model unchanged; compare row by row with Table 6.2.

| ID    | Parts | Train | Exact (%)      | PR-AUC (%)     | ID   | Parts | Train | Exact (%)      | PR-AUC (%)     |
|-------|-------|-------|----------------|----------------|------|-------|-------|----------------|----------------|
| L1    | 97    | 62    | 46.5 $\pm$ 8.3 | 64.1 $\pm$ 2.2 | L4   | 112   | 71    | 61.0 $\pm$ 3.7 | 61.7 $\pm$ 4.5 |
| L2    | 97    | 62    | 56.1 $\pm$ 5.0 | 61.6 $\pm$ 3.4 | L4+M | 112   | 71    | 61.1 $\pm$ 4.4 | 62.3 $\pm$ 4.0 |
| L2+T  | 97    | 62    | 60.5 $\pm$ 1.2 | 61.6 $\pm$ 3.4 | L5   | 160   | 102   | 66.4 $\pm$ 2.9 | 63.6 $\pm$ 1.9 |
| L2+TC | 97    | 62    | 56.6 $\pm$ 3.2 | 62.9 $\pm$ 2.7 | L6   | 182   | 116   | 68.9 $\pm$ 3.0 | 64.2 $\pm$ 3.1 |
| L3    | 92    | 58    | 60.5 $\pm$ 5.9 | 63.2 $\pm$ 3.8 | L7   | 182   | 116   | 72.6 $\pm$ 3.8 | 64.3 $\pm$ 3.4 |
| L3+M  | 92    | 58    | 58.3 $\pm$ 7.1 | 62.9 $\pm$ 3.5 | L8   | 182   | 116   | 73.1 $\pm$ 4.4 | 64.0 $\pm$ 3.2 |

Cumulative L1  $\rightarrow$  L8: 46.5  $\rightarrow$  73.1 (+26.6), against 51.2  $\rightarrow$  76.3 (+25.2) with the holistic features

**Table B.2:** Decision-layer alternatives on identical models and probabilities, exact recognition in percent. Rounds 1–2: L3+M models, geometry+holistic, 92 parts. Round 3 (3 August 2026): production feature set, v9 labels, about 145 parts per fold before the inner threshold hold-out; the two judges score every candidate partition on 20 structural features and are trained by nested cross-fitting. The oracles use the ground truth and are upper bounds.

| Method                               | Idea                                                                 | Rounds 1–2  | Round 3                            |
|--------------------------------------|----------------------------------------------------------------------|-------------|------------------------------------|
| Global $\tau$ + connected components | baseline                                                             | 58.3        | 75.37 $\pm$ 2.83                   |
| Partition ranking                    | learned judge, ranking loss                                          | –           | <b>75.96 <math>\pm</math> 4.36</b> |
| Partition regression                 | same features, regression loss                                       | –           | 74.82                              |
| Objective-picked $\tau$ per part     | maximum net-evidence score                                           | 55.2        | 74.79                              |
| Correlation clustering [21]          | net-evidence merging                                                 | 54.7        | 73.36                              |
| Learned $\tau$ per part              | regressor on part statistics; 15 and 74 parts per fold in rounds 1–2 | 56.7 / 56.9 | 72.83                              |
| Quantile $\tau$ per part             | per-part probability quantile                                        | 53.4        | 69.50                              |
| Stability-picked $\tau$ per part     | widest stable plateau                                                | 44.2        | 60.49                              |
| Per-part oracle $\tau$               | upper bound                                                          | 70.6        | 83.61                              |
| Correlation-clustering oracle        | upper bound                                                          | 70.4        | 82.33                              |

Four conclusions came out, in order. The partition algorithm is not the bottleneck: the correlation-clustering oracle matches the  $\tau$  oracle within 0.3 points in rounds 1–2 and 1.3 in round 3, so I did not pursue lifted multicuts [22]. The per-part headroom is not predictable from probability statistics: no heuristic or learned selector I tried predicts it, including a cross-fitted regressor with 74 parts per fold. The learned partition-ranking judge is the only positive result, +0.6 with 51 wins, 38 losses and 93 ties over the 182 parts. Finally, the oracle itself overfits, since it picks the best of 17 thresholds using the test labels on about 30 groups per part. The genuinely extractable decision-layer headroom is therefore 1 to 3 points, to be claimed by better calibration rather than by machinery on top of a miscalibrated ranker.

## B.4 Feature families

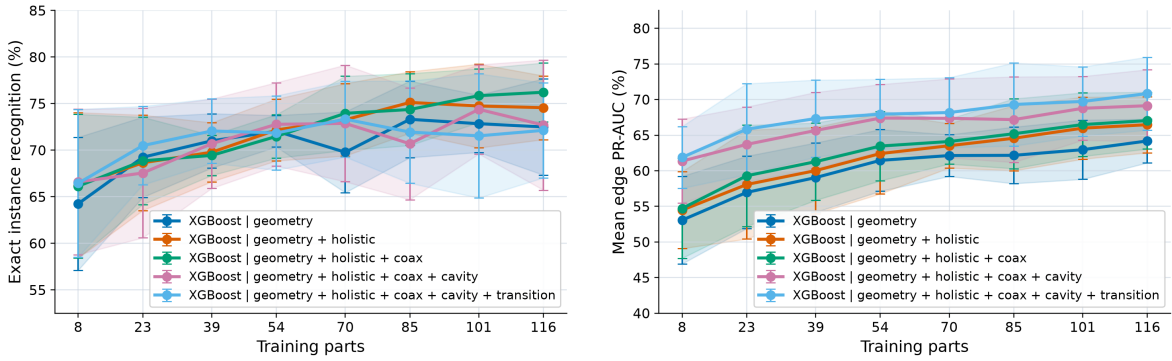
Table B.3 repeats the family comparison of rows F1 and F2 of Table 6.2 as one nested sweep on v9 labels and is the evidence for the production column of Table 5.1: ranking improves monotonically as families are added while exact recognition peaks at +coax, and Figure B.2 shows the ordering to be stable from about forty training parts, so it is not a small-sample

artefact. Table B.4 settles whether the loss is a thresholding problem: under a perfect per-part threshold the richer set is still 3.1 points worse, 81.1 against 84.2, so the ranking gains fall on pairs that do not decide partitions, and cavity and transition are dropped.

**Table B.3:** Cumulative feature families, **Table B.4:** Per-part-oracle test, 182 parts, exact recognition: 182-part curated corpus, 116 training parts, a perfect per-part threshold does not rescue the richer set. v9 labels.

| Feature set | Exact (%)      | PR-AUC (%)     |
|-------------|----------------|----------------|
| geometry    | 72.5 $\pm$ 5.2 | 64.2 $\pm$ 3.1 |
| +holistic   | 74.6 $\pm$ 3.4 | 66.5 $\pm$ 3.9 |
| +coax       | 76.2 $\pm$ 3.2 | 67.0 $\pm$ 4.0 |
| +cavity     | 72.7 $\pm$ 7.0 | 69.1 $\pm$ 5.1 |
| +transition | 72.1 $\pm$ 5.1 | 70.8 $\pm$ 5.1 |

| Feature set            | global $\tau$  | corr. clust. | oracle $\tau$  |
|------------------------|----------------|--------------|----------------|
| geometry+holistic+coax | 75.9 $\pm$ 2.8 | 73.8         | 84.2 $\pm$ 3.5 |
| +cavity+transition     | 71.9 $\pm$ 4.9 | 73.1         | 81.1 $\pm$ 5.6 |



**Figure B.2:** The capstone ablation as a sample-efficiency sweep over the five nested feature sets (v9 labels, 182 parts): exact recognition (left) and edge PR-AUC (right) against training parts per fold. On ranking the five sets separate in the order they were added and stay separated; on exact recognition the two rejected families fall below +coax from about forty training parts on, so the crossover is not a small-sample artefact. At 116 parts +coax ends highest on exact recognition (76.2) and the two richer sets end three to four points below it; bands  $\pm 1$  SD across folds.

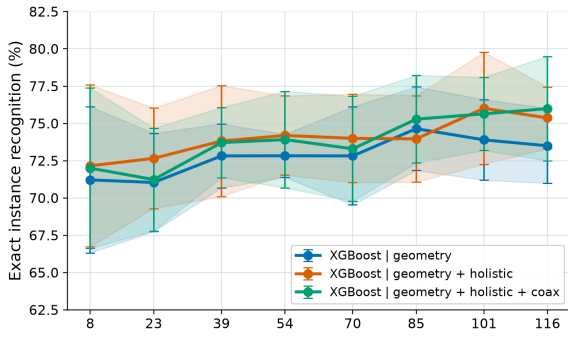
## B.5 Controlled data experiments

Two external corpora were appended to every fold’s training set: 97 simple generated parts of Zero-to-CAD and 102 complex real parts of the team’s older foxcad corpus (selected from 309; median 114 faces, up to 1,830; the 101 parts of the production pool plus one variant duplicate that the pool excludes). The curated test folds stayed fixed, so any gap is the effect of the added data alone (Table B.5, Figure B.3). The gain is +5.9 to +7.0 at eight curated training parts and within noise at 116 curated training parts. Under E1 the fold spread narrows by more than a third, from  $\pm 3.2$  to  $\pm 2.0$ . Beyond the plateau, volume therefore buys stability, not accuracy, which is the measurement behind the volume claim of Sections 6.2 and 7.2. A model trained on the blend of the curated and the generated parts (D3, 279 parts) scored about 75 on the real parts and about 89 on the generated ones, so a blended figure is the flattering one; the generated

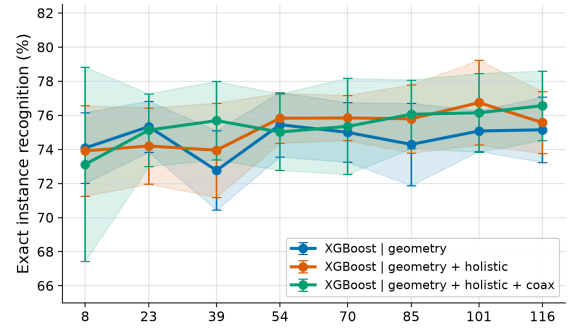
parts were left out of the 283-part production pool, and Section 6.2 quotes the pool apart from the curated parts.

**Table B.5:** D1 and E1: exact recognition on identical curated test folds, without and with external parts added to training only. Both sweeps covered all eight sizes; the Zero-to-CAD column is tabulated at the four logged ones, and panel (a) of Figure B.3 shows the full curve.

| Curated train parts | curated only   | +97 Zero-to-CAD, easy (D1) | +102 foxcad, hard (E1) |
|---------------------|----------------|----------------------------|------------------------|
| 8                   | 66.1 $\pm$ 7.7 | 72.0 $\pm$ 5.4 (+5.9)      | 73.1 $\pm$ 5.7 (+7.0)  |
| 23                  | 68.9 $\pm$ 4.7 | –                          | 75.1 $\pm$ 2.1 (+6.2)  |
| 39                  | 69.4 $\pm$ 2.2 | 73.7 $\pm$ 2.3 (+4.3)      | 75.7 $\pm$ 2.3 (+6.3)  |
| 54                  | 71.5 $\pm$ 2.3 | –                          | 75.0 $\pm$ 2.2 (+3.5)  |
| 70                  | 74.0 $\pm$ 4.0 | –                          | 75.4 $\pm$ 2.8 (+1.4)  |
| 85                  | 74.4 $\pm$ 3.9 | 75.3 $\pm$ 2.9 (+0.9)      | 76.1 $\pm$ 2.0 (+1.7)  |
| 101                 | 75.9 $\pm$ 2.9 | –                          | 76.2 $\pm$ 2.3 (+0.3)  |
| 116 (full)          | 76.2 $\pm$ 3.2 | 76.0 $\pm$ 3.5 (–0.2)      | 76.6 $\pm$ 2.0 (+0.4)  |



(a) D1: +97 easy generated parts, training only



(b) E1: +102 hard real parts, training only

**Figure B.3:** The two fixed-test sweeps, exact recognition against curated training parts per fold (8 to 116), three feature sets, bands  $\pm 1$  SD, external parts present at every point. Both sources lift the low-data end and both converge on the same in-domain ceiling of about 76 to 77; the easy parts still add about 4 points at 39 curated parts but under 1 by 85, whereas the hard parts still add about 2 there; both narrow the bands at the low-data end, and only the hard parts narrow them at 116 curated parts.

## B.6 Conventions

The census behind the first row of Table 6.3 gave every pair of adjacent faces a geometric signature and measured, over the corpus with the v8 labels, how often the two faces are labelled as one instance (Table B.6).

The A/B of the second row, designed by my supervisor, hand-labels the same 20 parts twice from the v12 groups with exactly one rule changed. Variant A merges every chamfer into the feature it breaks into; variant B makes chamfer rings their own instances unless they are integral to the feature’s shape. Both are trained identically on byte-identical folds with 13 training parts per fold (Table B.7), and Table B.8 confirms that the two labellings differ only in edge-break ownership. Two objections stand against the size of the +9.4: variant B also yields more and

**Table B.6:** Label consistency by interface category: the probability that two faces in the given relation are labelled as one instance, pooled over the corpus and dispersed across parts; the last column is the share of pairs no part-agnostic model can resolve, because identical configurations carry opposite labels across parts. The across-part range was recorded for the coaxial category only (–: not recorded).

| Interface category        | Pairs  | Parts | Pooled $P(\text{same})$ | p10–p90 across parts | Inconsistent |
|---------------------------|--------|-------|-------------------------|----------------------|--------------|
| adjacent + coaxial        | 20,916 | 168   | 0.342                   | 0.000–0.551          | 10.9 %       |
| smooth-adjacent (tangent) | 18,611 | 133   | 0.744                   | –                    | 7.7 %        |
| concave-adjacent          | 9,757  | 126   | 0.775                   | –                    | 3.9 %        |

simpler groups, which flatters an exact-match metric, and both labellings started from the v12 groups, which already followed rule B.

The third row of Table 6.3 is the programmatic harmonisation of the 283-part pool. I first tried to relabel a 40-part review set by hand, but abandoned it after about fifteen parts:  $P(\text{same})$  ranged from 0.12 to 0.70 on the review set, and many parts contradicted themselves. Instead, `apply_convention.py` applies the corpus-majority rule of each face-type category to every coaxial-adjacent pair. It cuts rule-different pairs inside a group and merges rule-same pairs across groups, deterministically and idempotently (Table B.9). On the most frequent coaxial interface, a cylinder against a plane (19,869 of the pool’s 34,845 coaxial adjacent pairs), the pool’s majority keeps a floor or shoulder apart from its bore ( $P(\text{same}) = 0.20$ ). The labellers of the newer 134-part bundle keep them together in 86 % of pairs. The two corpora follow inverted rules on that interface, which is one reason I never compare their scores.

**Table B.7:** Chamfer-convention A/B, per fold and in aggregate: exact recognition, then ranking quality and the selected thresholds. The whole gap sits in folds 1 and 4, where A collapses; in the other three folds the variants differ by at most 0.8 points either way.

| Exact recognition (%)                  | fold 1                      | fold 2 | fold 3 | fold 4      | fold 5 | mean               |
|----------------------------------------|-----------------------------|--------|--------|-------------|--------|--------------------|
| A: chamfers merged into their feature  | 44.2                        | 54.0   | 72.0   | 48.2        | 85.7   | 60.8 ± 17.5        |
| B: chamfers individual unless integral | <b>65.7</b>                 | 54.2   | 71.6   | <b>74.6</b> | 84.9   | <b>70.2 ± 11.3</b> |
| Edge PR-AUC (%)                        | A 62.1 ± 9.6 / B 63.6 ± 6.6 |        |        |             |        |                    |
| Selected $\tau$ across folds           | A 0.35–0.60 / B 0.50–0.65   |        |        |             |        |                    |

**Table B.8:** Label census of the two variants after extraction,  $P(\text{same})$  per interface category: only edge-break ownership differs. The 0.37 under B is the integral-chamfer share, concentrated in mould and fixture parts with bottom chamfers.

| Interface category                     | A    | B    |
|----------------------------------------|------|------|
| chamfer–hole                           | 0.91 | 0.37 |
| cylinder–torus (entry rounds, fillets) | 0.87 | 0.62 |
| cylinder–plane                         | 0.26 | 0.25 |

**Table B.9:** Programmatic harmonisation of the 283-part pool (labeled\_v11  $\rightarrow$  labeled\_v12\_conv): no gain in the headline score, a quarter less fold spread, and a threshold tightened into a narrow band.

| Quantity                                              | Value                            |
|-------------------------------------------------------|----------------------------------|
| Cuts applied / derived cuts                           | 6,442 / 7,041                    |
| Merges applied / skipped (residual violations)        | 873 / 175                        |
| Faces that changed group                              | <b>19,286</b>                    |
| Stock-strip instances split off / islands re-attached | 49 / 112                         |
| L9 before harmonisation                               | $79.07 \pm 3.53$ exact           |
| L9 after harmonisation                                | $78.87 \pm 2.69$ exact           |
| Fold spread                                           | $-24\%$                          |
| Per-fold $\tau$ after harmonisation                   | 0.75 / 0.75 / 0.65 / 0.65 / 0.70 |
| Per-fold exact after harmonisation                    | 82.2 / 77.3 / 75.4 / 80.7 / 78.7 |

## B.7 Model settings and the export re-tune

Table B.10 is the record behind the production configuration and the export re-tune of Section 6.2, and behind the gap of about 35 points between the training and the test parts of the same folds that Section 7.1 quotes.

**Table B.10:** Candidate generation, classifier, threshold protocol and the export re-tune of approach 2. Bundle results are given as v0.0.1 / v0.0.2 and are not comparable with the curated corpus; the effect of removing the calibration was never measured in isolation.

| Item                                                            | Value                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Candidate generation</i>                                     |                                                                                                                                                                                                                                                                                                                                         |
| Pair set                                                        | every CAD adjacency; $k$ nearest neighbours in centroid space ( $k = 64$ ), among faces of the same surface type ( $k = 8$ ) and in a normalised descriptor space ( $k = 32$ ); mirror-symmetry candidates; all pairs among the 32 largest faces                                                                                        |
| On the v0.0.2 bundle                                            | 134 parts, 731,617 candidate pairs, 43,001 sharing an edge, base rate 0.202; a perfect classifier restricted to the candidates still reaches 100 exact on 133 of the 134 parts                                                                                                                                                          |
| <i>Production classifier and threshold</i>                      |                                                                                                                                                                                                                                                                                                                                         |
| XGBoost                                                         | 200 trees, depth 6, learning rate 0.1, histogram method, positive class weighted by the negative-to-positive ratio, PR-AUC as training metric; isotonic calibrator fitted by three-fold cross-validation                                                                                                                                |
| Threshold $\tau$                                                | grid of 17 values from 0.10 to 0.90; chosen on an inner 20 % hold-out of the training parts (reshuffled with seed 42 + fold + 1) by maximising exact recognition, pairwise grouping F1 as tie-break; single-class parts excluded from the ranking metric                                                                                |
| <i>Export re-tune for the compiled path</i>                     |                                                                                                                                                                                                                                                                                                                                         |
| Changes                                                         | calibrator removed, so that <code>t12cgen</code> compiles the plain booster; positive pairs weighted by $1/\#\text{pairs}$ in their instance; 600 trees, depth 8, learning rate 0.05, row and column subsampling 0.8, minimum child weight 5; $\tau$ fixed at 0.88, the middle of a 0.84–0.90 plateau, from a two-fit-per-fold protocol |
| Effect of the re-tune on v0.0.1 (baseline already uncalibrated) | 86 parts, re-tuning run: exact 56.4 $\rightarrow$ 62.4; multi-face exact 34.3 $\rightarrow$ 50.3                                                                                                                                                                                                                                        |
| At $\tau = 0.88$                                                | cross-validation 61.8 $\pm$ 7.3 / 60.6 $\pm$ 8.0 from the exported fit, a separate run of the same recipe (the 62.4 above is the re-tune’s own run); on the training parts of the same folds 96.2 / 95.1                                                                                                                                |
| Trivial rules                                                   | every face its own instance 44.0 / 44.4; every adjacency merged 0.0; faces sharing a concave region merged 52.1 / 55.3                                                                                                                                                                                                                  |
| <i>Deployment contract</i>                                      |                                                                                                                                                                                                                                                                                                                                         |
| Feature order                                                   | frozen into an append-only 36-name contract with the C++ side, the 33 production features plus the three cavity features the concave-region rule reuses; booster exported as JSON with a metadata file                                                                                                                                  |
| Not built                                                       | the native half: no export module, no native directory, no golden-vector parity tests, no <code>t12cgen</code> invocation; two rounding sites in candidate generation are known not to reproduce across the two languages                                                                                                               |

## C Approach 3: what the agent is told and how it was tuned

This appendix extends Sections 5.4 and 6.3 with the text the model actually reads, the record of the configurations tried before the reported one was frozen, and the held-out run broken down further than Table 6.5.

### C.1 What the agent is told

The system prompt (`system_part_v7.md`, frozen on 24 September 2026) has the five parts of Table C.1, followed by the tool list and a discipline block: judge every spot and accept the rest as proposed, repeated features are separate instances, at most the budgeted number of tool calls, and only `finish_part` counts as an answer. The conventions (`conventions_agent_v3.md`) are substituted into it. Their opening is reproduced below with markdown emphasis removed and arrows in ASCII. Version 3 differs from version 2 in one rule, D2. Its earlier wording, that an outside round belongs to the wall and not to the top face, contradicted the labels in 77 % of cases and made the model split welds wrongly. D2 now states the measured behaviour.

**Table C.1:** The five parts of the system prompt of the frozen configuration.

| Part        | What it says                                                                                                                                                                                                                                                                                  |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task        | group every face of a grey part into the instances a CAM programmer would machine in one step, each face in exactly one instance and a lone face its own instance; name each instance; then <code>finish_part</code>                                                                          |
| Walk        | propose the next rule group with its spots; no spots: accept; spots: judge each, then accept, which cuts the links judged apart and re-splits the group into its connected pieces                                                                                                             |
| Spots       | <i>bridge</i> : is face X, which holds several pieces together, the same feature as this piece (together if the piece is cut into or grows from X, apart if X is stock, a land or another feature it merely touches); <i>flat</i> : does a flat stay with the outside rounds it is tangent to |
| Confidence  | 1 to 5; a verdict is applied at 4 or 5. Below that, the spot's default applies: the labellers' majority for that kind of bridge, and together for flats                                                                                                                                       |
| Conventions | the numbered rules of <code>conventions_agent_v3.md</code> , sections A to H (contours, pockets and bosses, holes, chamfers and rounds, patterns, freeform, how to commit, sheet metal), each with the rate at which the labellers followed it, measured on the development parts only        |

An instance is the set of faces one machining step produces, as the CAM programmer who labeled these parts saw it: one pocket with its walls, one hole (with everything drilled, bored and chamfered on the same axis), one boss with the face it stands on, one tangent chain of contour walls, one lone top face. A face belongs to exactly one instance.

1. CONCAVE sharp edge (angle < 180) -> same instance: 95 %. A wall and the floor it rises from are cut together; so are the walls of a boss, pad, rib or pin and the face they stand on; so is a wall and its floor fillet.

Exceptions are rare (a hole through a pocket floor, rule C3).

2. TANGENT edge (angle = 180, G1/G2) -> same instance: 87 %. Blend chains, corner rounds, slot ends, the round strips along a contour, a face that continues smoothly into a rounded wall. Weaker cases: a small round band (cylindrical or toroidal) tangent to a large planar face joins it only ~60-70 % of the time; a blend band tangent to a cylinder ~40 %.
3. CONVEX sharp edge (angle > 180) -> different instances: 89 %. This includes sharp corners between two side walls of a contour (only 27 % same) and a narrow planar band at a chamfer angle next to a planar face (only 6 % same). The reliable exceptions: a conical or cylindrical band at the mouth of a hole belongs to the hole (89 %); the shoulders and bores of one coaxial hole stack belong together (rule C2); a chamfer at the end of a boss or shaft belongs to that boss (rule D1).

Default when in doubt: concave and tangent neighbours in, convex neighbours out.

## C.2 Design ledger on the development parts

Table C.2 is the record behind the development rows of Table 6.4 and lists the configurations in the order they were tried. Every configuration was scored on the same 24 development parts (295 instances) by the same function, and the reported configuration was frozen before its own held-out run. The picture arms were probed offline on the 124 bridge questions of the 78 development parts, three runs per arm, so their block has its own four columns (see the caption). Every picture design cost 5 to 8 points per question against text alone, which is the finding quoted in Section 6.3. Text alone is itself no better than the class default per question (61.0 against 62.1); it pays only through its confident answers (65.0) and the gated walk (67.3 against 66.8), which is why verdicts are gated. I also tried worked examples in the prompt on an earlier per-face version of the task; they gave no gain once the example parts were removed from the sample, so I did not carry them forward.

Two repeat runs measure run-to-run variation: the v5 walk produced the identical partition on 21 of 24 parts with 12 of 152 verdicts flipped, and a repeat of v7 reproduced its aggregate exactly (74.0 exact, 62.8 multi, 68.0 balanced), where the free-form agent had scored 6 of 11 and 10 of 11 instances on the same part in two runs.

## C.3 The held-out run by complexity class

Table C.3 breaks the frozen held-out run of Table 6.4 (119 parts, 1,325 instances, 91.7 credits in all, 2.15 hours of agent time across two servers; 32 calls and no screenshot per part) down by the bundle's complexity classes, as Table 6.5 does by part size: the block-and-cylinder class is close to solved, and the freeform (class 8), intersecting (4) and elevated-complexity (7) classes, small as they are, are where both the rules and the verdicts run out; the relatively freeform class 1 is not among them.

Table C.4 lists the ten hardest held-out parts, identified by rank only since their file names are confidential: four of the worst five predict far fewer instances than the labels hold, at pairwise

**Table C.2:** Design ledger of approach 3 on the 24 development parts, in the order tried, in percent; v1 to v7 number the system-prompt configurations (v7 is `system_part_v7.md` of Appendix C.1), not the conventions file, which is at version 3 in v7; calls and credits per part are from the run reports (one credit is one cent; the free-form agent took 0.8 screenshots per part, every walk none). Picture block: per-question accuracy, accuracy of the answers given at confidence 4 or 5, balanced score of the walk replayed with those answers, and the share of answers at confidence 4 or 5. Bold: best of each block.

| Configuration                                                                                 | Exact       | Multi       | Balanced    | Calls | Credits |
|-----------------------------------------------------------------------------------------------|-------------|-------------|-------------|-------|---------|
| classifier of approach 2 (out-of-fold)                                                        | 50.9        | 58.6        | 54.5        | –     | –       |
| rules alone (concave-tangent, near-tangent, chamfer, stack)                                   | 66.9        | 54.1        | 59.8        | –     | 0       |
| agent from blank, free-form (v1)                                                              | 61.5        | 44.3        | 51.5        | 48    | 1.34    |
| agent repairing the concave-tangent draft (v3)                                                | 64.0        | 49.6        | 55.8        | 15    | 0.57    |
| walk: flat and weak-link spots (v5; a weak link is a weld the rules make with low confidence) | 69.1        | 54.6        | 61.0        | 39    | 1.04    |
| walk: flat spots only, confidence gate (v6)                                                   | 66.5        | 52.6        | 58.7        | 35    | 0.86    |
| walk: bridge and flat spots, gate, re-split (v7, frozen)                                      | <b>74.0</b> | <b>62.8</b> | <b>68.0</b> | 35    | 0.90    |
| v7 without flat spots                                                                         | 72.0        | 59.6        | 65.2        | 33    | 0.85    |

*Pictures at bridge spots: offline probe, 124 questions of the 78 development parts, three runs per arm*

| Arm                                                      | per question | right if confident | balanced, walk | confident share |
|----------------------------------------------------------|--------------|--------------------|----------------|-----------------|
| class default, no model                                  | <b>62.1</b>  | –                  | 66.8           | –               |
| text only                                                | 61.0         | <b>65.0</b>        | <b>67.3</b>    | 64              |
| plain picture, labels only                               | 54.3         | 56.2               | 66.0           | 84              |
| X and the asked piece in blue                            | 54.0         | 55.4               | 66.0           | 79              |
| pieces in their own hues, neutral paragraph              | 55.6         | 56.6               | 66.9           | 78              |
| pieces in their own hues, paragraph naming feature kinds | 52.7         | 54.0               | 66.7           | 80              |

**Table C.3:** The frozen held-out run by the bundle’s complexity class: exact recognition, its multi-face variant, their harmonic mean, pairwise accuracy and the number of parts with every instance exact. Classes with five parts or fewer are indicative only.

| Class                                 | Parts | Exact (%) | Multi (%) | Balanced (%) | Pair acc. (%) | Perfect parts |
|---------------------------------------|-------|-----------|-----------|--------------|---------------|---------------|
| 0 block and cylinder                  | 24    | 89.9      | 82.9      | 86.2         | 94.7          | 10            |
| 1 relatively freeform                 | 14    | 79.7      | 67.8      | 73.3         | 95.5          | 5             |
| 2 large part                          | 1     | 57.1      | 47.1      | 51.6         | 98.4          | 0             |
| 3 profile                             | 33    | 75.1      | 62.5      | 68.3         | 93.8          | 4             |
| 4 intersecting                        | 4     | 61.3      | 50.0      | 55.1         | 83.6          | 0             |
| 5 standard                            | 3     | 70.6      | 68.1      | 69.3         | 97.6          | 0             |
| 6 profile, intersecting, disconnected | 27    | 74.3      | 64.1      | 68.8         | 91.6          | 3             |
| 7 elevated complexity                 | 5     | 69.0      | 59.3      | 63.8         | 87.9          | 0             |
| 8 freeform                            | 5     | 56.2      | 54.8      | 55.5         | 75.3          | 1             |
| 9 disconnected                        | 3     | 71.8      | 67.7      | 69.7         | 95.9          | 0             |
| all classes                           | 119   | 76.6      | 66.6      | 71.2         | 92.5          | 23            |

accuracy between 19 % and 57 %, so the dominant failure on the hardest parts is a group the walk welded and no verdict cut apart; the other six sit at pairwise accuracy of 69 % or more, where the errors are single wrong welds or over-splits rather than a collapsed part.

**Table C.4:** The ten hardest held-out parts of the frozen run, by exact recognition: faces, labelled and predicted instance counts, exact recognition and pairwise accuracy; every part was fully assigned.

| Rank | Faces | Labelled instances | Predicted | Exact (%) | Pair acc. (%) |
|------|-------|--------------------|-----------|-----------|---------------|
| 1    | 29    | 6                  | 1         | 0         | 19            |
| 2    | 54    | 12                 | 2         | 8         | 33            |
| 3    | 30    | 9                  | 14        | 11        | 88            |
| 4    | 43    | 7                  | 2         | 14        | 57            |
| 5    | 61    | 9                  | 5         | 22        | 52            |
| 6    | 50    | 18                 | 12        | 28        | 69            |
| 7    | 24    | 16                 | 13        | 38        | 94            |
| 8    | 39    | 9                  | 12        | 44        | 91            |
| 9    | 58    | 8                  | 8         | 50        | 99            |
| 10   | 49    | 6                  | 11        | 50        | 95            |